

The simplified MITC4+ shell element and its performance in linear and nonlinear analysis

Hyung-Gyu Choi, Phill-Seung Lee^{*}

Department of Mechanical Engineering, Korea Advanced Institute of Science and Technology, 291 Daehak-ro, Yuseong-gu, Daejeon 34141, Republic of Korea

ARTICLE INFO

Keywords:

4-node shell finite element
Assumed strain
Finite element method
Mesh distortion
MITC method
Numerical integration

ABSTRACT

The recently developed improved MITC4+ shell element exhibits excellent convergence behavior in both bending- and membrane-dominated shell problems. Compared to the widely recognized MITC4 shell element, the membrane and in-plane shear locking were significantly alleviated. However, it is difficult to implement the assumed membrane strain field because of its complexity. In this study, a simplified MITC4+ shell element is developed. A simpler assumed membrane strain field that facilitates the extension to nonlinear formulations is proposed. To reduce the sensitivity of the element performance to mesh distortion, a geometry-dependent Gauss integration scheme is employed. The proposed element passes the patch, zero-energy mode, and isotropy tests. The performance of the simplified MITC4+ shell element is demonstrated using various linear and nonlinear benchmark problems.

1. Introduction

Numerous shell structures found in nature demonstrate excellent load-bearing capabilities. Shell structures are used extensively in various engineering fields [1–7], and the finite element method (FEM) has been predominantly used for shell structure analysis. In the practical finite element analysis of shells, the use of 4-node quadrilateral shell elements is preferred owing to their simplicity and efficiency [8–10].

Shell finite elements suffer from shear and membrane locking, which occur when finite element discretization cannot accurately represent the pure bending displacement fields [10–13]. Locking significantly deteriorates the solution accuracy, particularly when the shell thickness is small. Over the past few decades, numerous efforts have been made to overcome locking problems resulting in the development of effective locking treatment methods such as assumed strain, incompatible modes, and reduced integration [14–25].

The mixed interpolation of tensorial components (MITC) method has been successfully applied to alleviate transverse shear locking [26–33] leading to the development of a 4-node MITC shell element (MITC4). Most commercial FE software adopt the assumed transverse shear strains of the MITC4 shell element. Ko et al. developed MITC4+ shell elements in which membrane locking was reduced using the MITC method [10,34]. In addition, two-dimensional (2D) and three-dimensional (3D) MITC solid elements have been developed [35–38]. Their excellent

performance has been demonstrated by both linear and nonlinear analysis [9,39,40].

Ko et al. recently introduced the improved MITC4+ shell element in which the in-plane shear locking was also alleviated by adopting the assumed strain field of the 2D MITC solid element [37]. It exhibited the best predictive capability among the MITC4/MITC4+ shell elements in linear analysis. However, the improved MITC4+ shell element has a highly complex assumed membrane strain field which makes its implementation difficult. Extending this formulation to nonlinear analysis is a difficult task.

This study aims to provide a simplified assumed strain field for the improved MITC4+ shell element. Transverse shear, membrane, and in-plane shear locking are alleviated. First, a generalized assumed membrane strain field is introduced that encompasses the assumed membrane strain field of the improved MITC4+ shell element. Subsequently, a simplified assumed strain field is obtained by merging the tying points into the element center. To improve the membrane (in-plane) performance in distorted meshes further, a recently developed geometry-dependent Gauss integration scheme is adopted [38].

The formulation of the MITC4+ shell elements is briefly reviewed in the following sections. A simplified MITC4+ shell element is presented in detail. The performance of the simplified MITC4+ shell element is demonstrated through elementary numerical tests and several linear and nonlinear problems, considering both regular and distorted meshes.

^{*} Corresponding author.

E-mail address: phillseung@kaist.edu (P.-S. Lee).

<https://doi.org/10.1016/j.compstruc.2023.107177>

Received 24 April 2023; Accepted 10 September 2023

Available online 16 October 2023

0045-7949/© 2023 Elsevier Ltd. All rights reserved.

Table 1

Locking treatments of the 4-node quadrilateral MITC/MITC+ shell elements. The symbols '✓' and '-' mean that locking is treated or not, respectively.

Elements	Transverse shear locking	Membrane locking	In-plane shear locking
MITC4 1984, Dvorkin and Bathe [4]	✓	–	–
MITC4+ 2016, Ko et al. [10]	✓	✓	–
New MITC4+ 2017, Ko et al. [34]	✓	✓	–
Improved MITC4+ 2017, Ko et al. [37]	✓	✓	✓
Simplified MITC4+ (proposed in this study)	✓	✓	✓

2. MITC4+ shell elements

In this section, the formulation of the MITC4+ shell elements is briefly reviewed focusing on the locking treatments. The locking treatments for the 4-node quadrilateral MITC shell elements are listed in Table 1. All MITC elements adopt the assumed transverse shear strains proposed in Ref. [4]. In the MITC4+ shell element, membrane locking was alleviated to a certain extent [10]. The new and improved MITC4+

shell elements employed the assumed membrane strains proposed in Ref. [34]. In the improved MITC4+ shell element, in-plane shear locking was treated by using the assumed strains of the 2D-MITC4 element [37].

2.1. Geometry and displacement interpolations

The geometry interpolation of the continuum-mechanics-based 4-node quadrilateral shell finite element is given by [1,31]

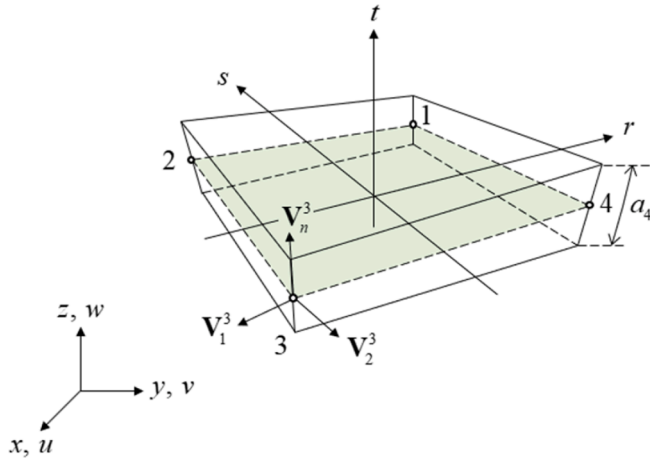


Fig. 1. Geometry of a 4-node quadrilateral shell element.

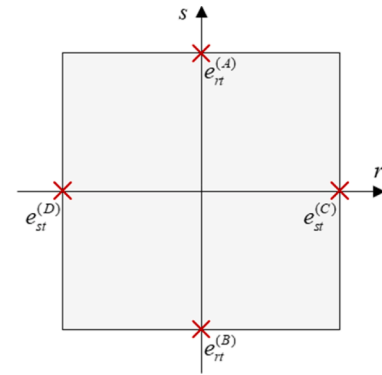


Fig. 3. Tying points (A)–(D) for the assumed transverse shear strain field and the corresponding strain components.

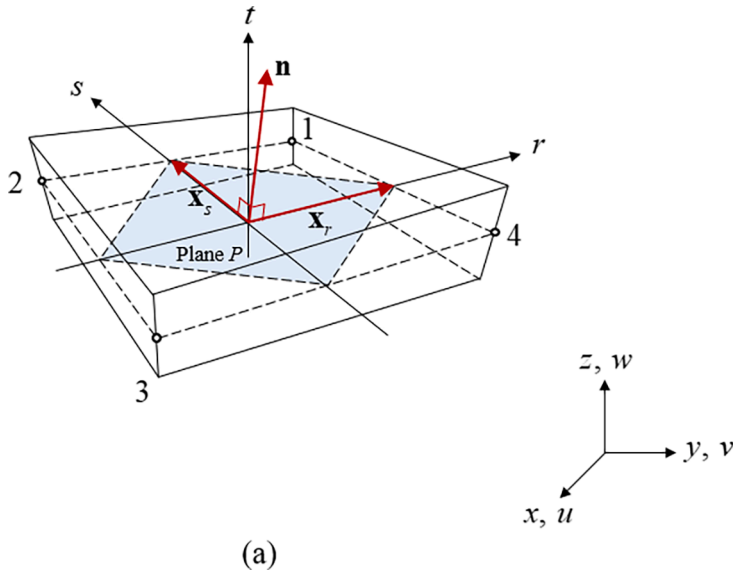


Fig. 2. Characteristic geometry vectors for a 4-node quadrilateral shell element. (a) Two vectors $\mathbf{x}_r$ and $\mathbf{x}_s$ in the plane P with a normal vector $\mathbf{n}$. (b) Distortion vector $\mathbf{x}_d$.

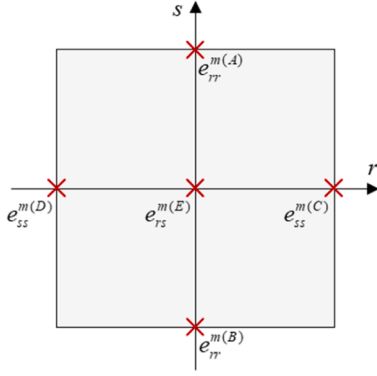


Fig. 4. Tying points (A)–(E) for the assumed membrane strain field and the corresponding strain components.

Table 2

Matrices $\mathbf{M}$ and $\mathbf{C}$ of the improved and simplified MITC4+ shell elements. Note that $g_j^i|_{(\cdot)}$ is g_j^i evaluated at a tying point $(\cdot)$, see Eqs. (26) and (27).

Improved MITC4+ (2017, Ko et al., [37])	Simplified MITC4+ (proposed in this study)
$\mathbf{M} = \lambda \begin{bmatrix} 1/\lambda + \sqrt{3}n_1s & \sqrt{3}n_2s & 2\sqrt{3}n_3s & n_3s & n_4s & n_1s/\sqrt{3} \\ \sqrt{3}m_2r & 1/\lambda + \sqrt{3}m_1r & 2\sqrt{3}m_5r & m_4r & m_3r & m_1r/\sqrt{3} \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$ $n_1 = \frac{1}{2}[(g_r^r _{(A)})^2 - (g_r^r _{(B)})^2]$ $n_2 = \frac{1}{2}[(g_r^s _{(A)})^2 - (g_r^s _{(B)})^2]$ $n_3 = \frac{1}{2}[(g_r^r _{(A)})^2 + (g_r^r _{(B)})^2]$ $n_4 = \frac{1}{2}[g_r^r _{(A)}g_r^s _{(A)} + g_r^r _{(B)}g_r^s _{(B)}]$ $n_5 = \frac{1}{2}[g_r^r _{(A)}g_r^s _{(A)} - g_r^r _{(B)}g_r^s _{(B)}]$ $m_1 = \frac{1}{2}[(g_s^s _{(C)})^2 - (g_s^s _{(D)})^2]$ $m_2 = \frac{1}{2}[(g_s^r _{(C)})^2 - (g_s^r _{(D)})^2]$ $m_3 = \frac{1}{2}[(g_s^s _{(C)})^2 + (g_s^s _{(D)})^2]$ $m_4 = \frac{1}{2}[g_s^r _{(C)}g_s^s _{(C)} + g_s^r _{(D)}g_s^s _{(D)}]$ $m_5 = \frac{1}{2}[g_s^r _{(C)}g_s^s _{(C)} - g_s^r _{(D)}g_s^s _{(D)}]$ $\mathbf{C} = \begin{bmatrix} 1/2 - a_A & 1/2 - a_B & -a_C & -a_D & -a_E \\ -a_A & -a_B & 1/2 - a_C & 1/2 - a_D & -a_E \\ 0 & 0 & 0 & 0 & 1 \\ 1/2 & -1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & -1/2 & 0 \\ a_A & a_B & a_C & a_D & a_E \end{bmatrix}$	$\mathbf{M} = \lambda \begin{bmatrix} 1/\lambda - 2as & 0 & -2\beta s & s & 0 \\ 0 & 1/\lambda - 2\beta r & -2ar & 0 & r \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$ $\mathbf{C} = \begin{bmatrix} 1/2 - a_A & 1/2 - a_B & -a_C & -a_D & -a_E \\ -a_A & -a_B & 1/2 - a_C & 1/2 - a_D & -a_E \\ 0 & 0 & 0 & 0 & 1 \\ 1/2 & -1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & -1/2 & 0 \end{bmatrix}$

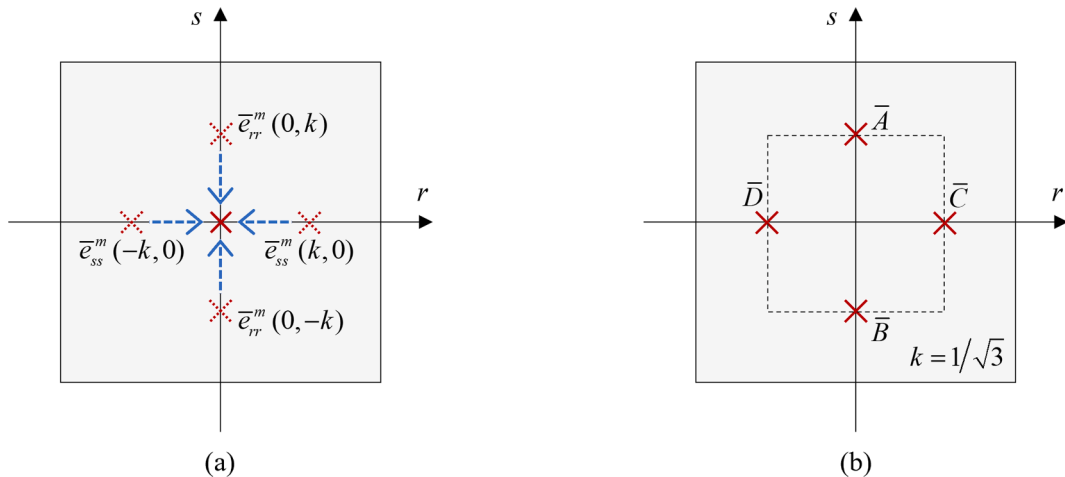


Fig. 5. Tying points for the assumed membrane strain field of (a) the simplified MITC4+ shell element and (b) the improved MITC4+ shell element.

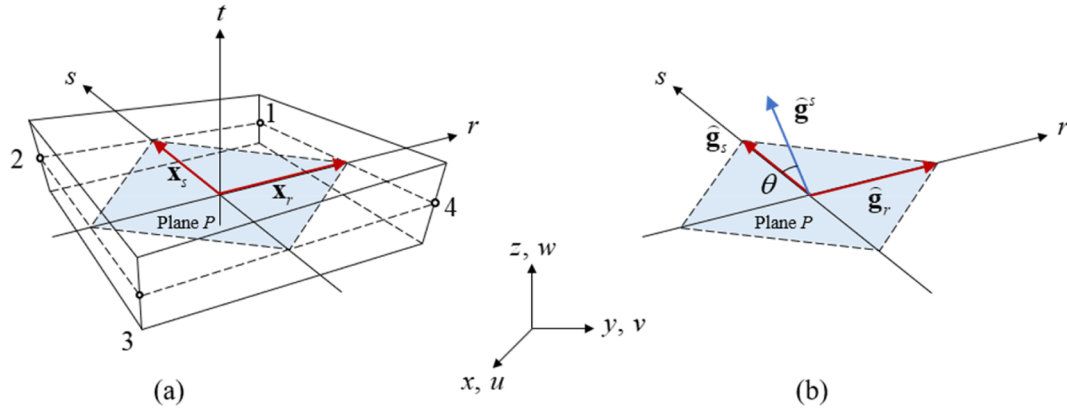


Fig. 6. Skewness of the 4-node quadrilateral shell element. (a) In-plane vectors $\mathbf{x}_r$ and $\mathbf{x}_s$. (b) Skew angle θ of the shell element.

$$\mathbf{u}(r, s, t) = \sum_{i=1}^4 h_i(r, s) \mathbf{u}_i + \frac{t}{2} \sum_{i=1}^4 a_i h_i(r, s) (-\alpha_i \mathbf{V}_2^i + \beta_i \mathbf{V}_1^i), \quad (5)$$

in which $\mathbf{u}_i$ is the displacement vector at node i , $\mathbf{V}_1^i$ and $\mathbf{V}_2^i$ are unit vectors orthogonal to the director vector $\mathbf{V}_n^i$ and to each other, and α_i and β_i denote the rotations of the director vectors about $\mathbf{V}_1^i$ and $\mathbf{V}_2^i$, respectively, at node i .

The displacement-based covariant strain components are given by [5,28]

$$e_{ij} = \frac{1}{2} (\mathbf{g}_i \cdot \mathbf{u}_{,j} + \mathbf{g}_j \cdot \mathbf{u}_{,i}), \quad (6)$$

in which

$$\mathbf{g}_i = \frac{\partial \mathbf{x}}{\partial r_i} \text{ and } \mathbf{u}_{,i} = \frac{\partial \mathbf{u}}{\partial r_i} \text{ with } r_1 = r, r_2 = s, r_3 = t. \quad (7)$$

Note that $e_{11} = e_{rr}$, $e_{22} = e_{ss}$, $e_{12} = e_{rs}$, $e_{13} = e_{rt}$ and $e_{23} = e_{st}$.

We can write the covariant in-plane strain in Eq. (6) as [10,30]

$$e_{ij} = e_{ij}^m + t e_{ij}^{b1} + t^2 e_{ij}^{b2} \text{ with } i, j = 1, 2, \quad (8)$$

$$e_{ij}^m = \frac{1}{2} \left(\frac{\partial \mathbf{x}_m}{\partial r_i} \frac{\partial \mathbf{u}_m}{\partial r_j} + \frac{\partial \mathbf{x}_m}{\partial r_j} \frac{\partial \mathbf{u}_m}{\partial r_i} \right), \quad (9)$$

Table 3

Eigenvalues of the stiffness matrices of the simplified MITC4+ shell element according to the adjusting parameter μ . Note that eigenvalues smaller than 10^{-15} are considered as zero.

Modes		$\mu = 1$	$\mu = 0.5$	$\mu = 0.1$	$\mu = 0$
7	BL1	7.2000E-07	4.5000E-08	7.1933E-11	0
8	BL2	7.2000E-07	4.5000E-08	7.1933E-11	0
9	BC1	9.9556E-07	9.9556E-07	9.9555E-07	9.9554E-07
10	BC2	1.1200E-06	1.1200E-06	1.1200E-06	1.1200E-06
11	BC3	2.0800E-06	2.0800E-06	2.0800E-06	2.0800E-06
12	TQ	5.6000E+01	3.5000E+00	5.6000E-03	0
13	BC4	5.0400E+02	3.1500E+01	5.0400E-02	0
14	ML1	6.4000E+02	4.0000E+01	6.4000E-02	0
15	ML2	6.4000E+02	4.0000E+01	6.4000E-02	0
16	SQ1	8.4000E+02	8.4000E+02	8.4000E+02	8.4000E+02
17	SQ2	8.4000E+02	8.4000E+02	8.4000E+02	8.4000E+02
18	MC1	1.3440E+03	1.3440E+03	1.3440E+03	1.3440E+03
19	MC2	1.3440E+03	1.3440E+03	1.3440E+03	1.3440E+03
20	MC3	2.4960E+03	2.4960E+03	2.4960E+03	2.4960E+03

MC: membrane modes with in-plane constant strain fields, ML: membrane modes with in-plane bending strain fields, BC: bending modes with constant bending strain fields, BL: bending modes with linear bending strain fields, TQ: in-plane twisting modes, SQ: transverse shear strain modes.

$$e_{ij}^{b1} = \frac{1}{2} \left(\frac{\partial \mathbf{x}_m}{\partial r_i} \frac{\partial \mathbf{u}_b}{\partial r_j} + \frac{\partial \mathbf{x}_m}{\partial r_j} \frac{\partial \mathbf{u}_b}{\partial r_i} + \frac{\partial \mathbf{x}_b}{\partial r_i} \frac{\partial \mathbf{u}_m}{\partial r_j} + \frac{\partial \mathbf{x}_b}{\partial r_j} \frac{\partial \mathbf{u}_m}{\partial r_i} \right), \quad (10)$$

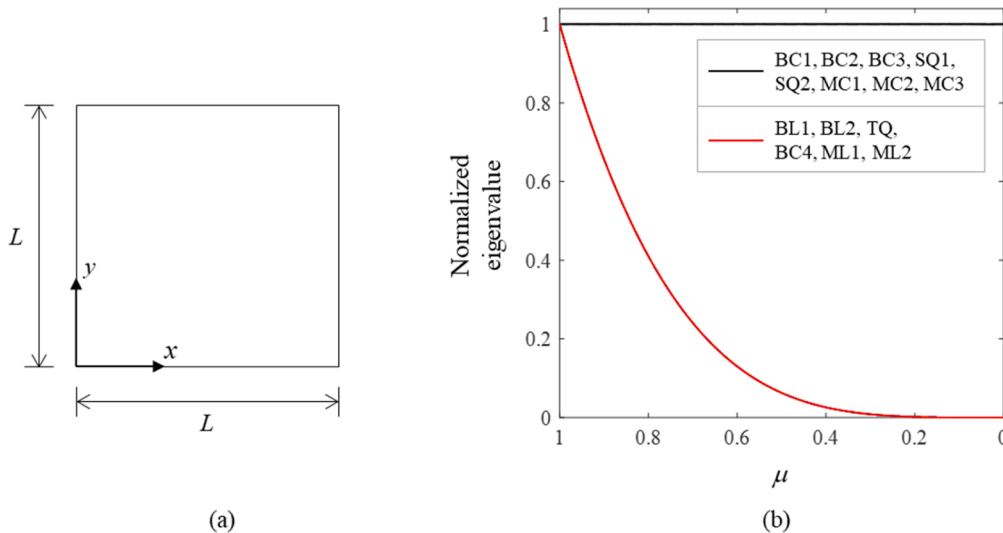


Fig. 7. Eigen analysis of the stiffness matrix. (a) Single simplified MITC4+ shell element ($L = 1$, $E = 1.7472 \times 10^7$ and $\nu = 0.3$). (b) Normalized eigenvalues according to the adjusting parameter μ .

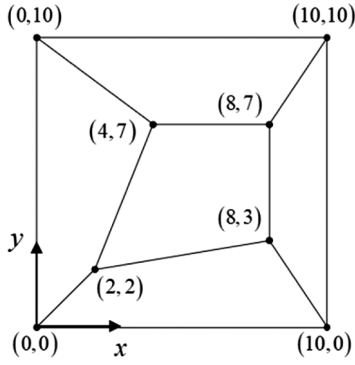


Fig. 8. Mesh geometry for the patch tests.

$$e_{ij}^{b2} = \frac{1}{2} \left(\frac{\partial \mathbf{x}_b}{\partial r_i} \frac{\partial \mathbf{u}_b}{\partial r_j} + \frac{\partial \mathbf{x}_b}{\partial r_j} \frac{\partial \mathbf{u}_b}{\partial r_i} \right), \quad (11)$$

with

$$\mathbf{x}_m = \sum_{i=1}^4 h_i(r, s) \mathbf{x}_i, \quad \mathbf{x}_b = \frac{1}{2} \sum_{i=1}^4 a_i h_i(r, s) \mathbf{V}_n^i, \quad (12)$$

$$\mathbf{u}_m = \sum_{i=1}^4 h_i(r, s) \mathbf{u}_i, \quad \mathbf{u}_b = \frac{1}{2} \sum_{i=1}^4 a_i h_i(r, s) (-\alpha_i \mathbf{V}_2^i + \beta_i \mathbf{V}_1^i). \quad (13)$$

In Eq. (8), the strain term e_{ij}^m is the covariant membrane strain at the shell mid surface ($t = 0$), and the remaining terms, including e_{ij}^{b1} and e_{ij}^{b2} , denote the covariant in-plane strain owing to bending.

For the derivatives in Eq. (9), the following relations are obtained as in Ref. [34]

$$\frac{\partial \mathbf{x}_m}{\partial r} = \mathbf{x}_r + s \mathbf{x}_d, \quad \frac{\partial \mathbf{x}_m}{\partial s} = \mathbf{x}_s + r \mathbf{x}_d, \quad \frac{\partial \mathbf{u}_m}{\partial r} = \mathbf{u}_r + s \mathbf{u}_d, \quad \frac{\partial \mathbf{u}_m}{\partial s} = \mathbf{u}_s + r \mathbf{u}_d \quad (14)$$

with

$$\mathbf{x}_r = \frac{1}{4} \sum_{i=1}^4 \xi_i \mathbf{x}_i, \quad \mathbf{x}_s = \frac{1}{4} \sum_{i=1}^4 \eta_i \mathbf{x}_i, \quad \mathbf{x}_d = \frac{1}{4} \sum_{i=1}^4 \xi_i \eta_i \mathbf{x}_i, \quad (15)$$

$$\mathbf{u}_r = \frac{1}{4} \sum_{i=1}^4 \xi_i \mathbf{u}_i, \quad \mathbf{u}_s = \frac{1}{4} \sum_{i=1}^4 \eta_i \mathbf{u}_i, \quad \mathbf{u}_d = \frac{1}{4} \sum_{i=1}^4 \xi_i \eta_i \mathbf{u}_i, \quad (16)$$

in which the vectors $\mathbf{x}_r$, $\mathbf{x}_s$ and $\mathbf{x}_d$ are the characteristic geometry vectors as shown in Fig. 2, and the corresponding displacement vectors are $\mathbf{u}_r$, $\mathbf{u}_s$ and $\mathbf{u}_d$, respectively. The two vectors $\mathbf{x}_r$ and $\mathbf{x}_s$ construct the plane P with the normal vector $\mathbf{n}$,

$$\mathbf{n} = \frac{\mathbf{x}_r \times \mathbf{x}_s}{|\mathbf{x}_r \times \mathbf{x}_s|}. \quad (17)$$

2.2. Assumed transverse shear strains

To reduce the transverse shear locking in the MITC4+ shell element, the assumed transverse shear strain field of the MITC4 shell element is adopted [34]. Using the four tying points (A)–(D) shown in Fig. 3, the following assumed strain field is obtained [4]:

$$\tilde{e}_{rt} = \frac{1}{2} (1 + s) e_{rt}^{(A)} + \frac{1}{2} (1 - s) e_{rt}^{(B)}, \quad (18)$$

$$\tilde{e}_{st} = \frac{1}{2} (1 + r) e_{st}^{(C)} + \frac{1}{2} (1 - r) e_{st}^{(D)}. \quad (19)$$

In the formulation of the MITC4 shell element, membrane locking is not treated. Therefore, the MITC4 element suffers from membrane locking in particular when distorted [10]. For this reason, the MITC4+ shell element was developed to alleviate the membrane locking.

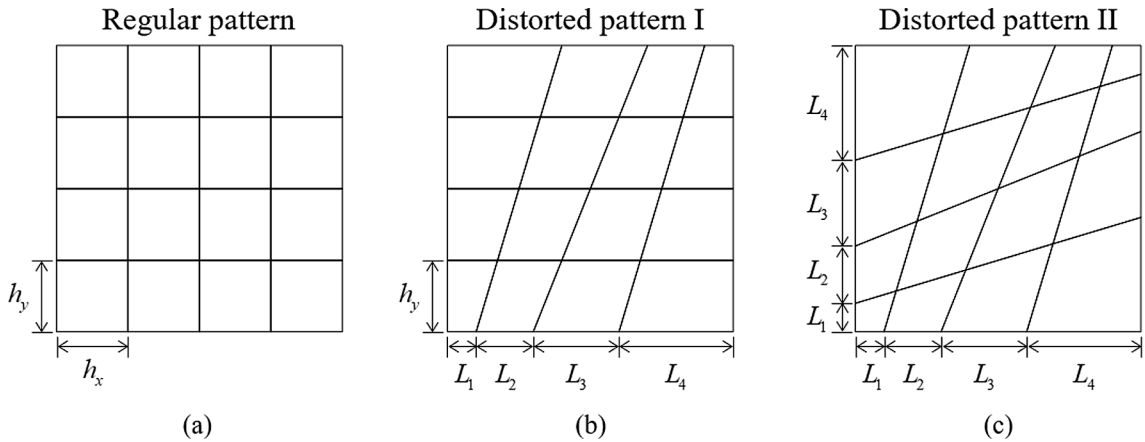
2.3. Assumed membrane strains

For a 4-node quadrilateral shell element, membrane locking occurs when the element geometry is warped [10]. To alleviate membrane locking using tying points (A)–(E), as shown in Fig. 4, the assumed membrane strain field of the new MITC4+ shell element is constructed as follows [34]:

$$\tilde{e}_{rr}^m = \frac{1}{2} (1 + s + 2(s^2 - 1)a_A) e_{rr}^{m(A)} + \frac{1}{2} (1 - s + 2(s^2 - 1)a_B) e_{rr}^{m(B)} + (s^2 - 1)a_C e_{ss}^{m(C)} + (s^2 - 1)a_D e_{ss}^{m(D)} + (s^2 - 1)a_E e_{rs}^{m(E)}, \quad (20)$$

$$\tilde{e}_{ss}^m = (r^2 - 1)e_{rr}^{m(A)} + (r^2 - 1)e_{rr}^{m(B)} + \frac{1}{2} (1 + r + 2(r^2 - 1)a_C) e_{ss}^{m(C)} + \frac{1}{2} (1 - r + 2(r^2 - 1)a_D) e_{ss}^{m(D)} + (r^2 - 1)a_E e_{rs}^{m(E)}, \quad (21)$$

$$\tilde{e}_{rs}^m = \frac{1}{4} (r + 4rsa_A) e_{rr}^{m(A)} + \frac{1}{4} (-r + 4rsa_B) e_{rr}^{m(B)} + \frac{1}{4} (s + 4rsac) e_{ss}^{m(C)} + \frac{1}{4} (-s + 4rsad) e_{ss}^{m(D)} + (1 + rsa_E) e_{rs}^{m(E)} \quad (22)$$

Fig. 9. Three mesh patterns with $N = 4$. (a) Regular pattern, (b) Distorted pattern I, (c) Distorted pattern II.

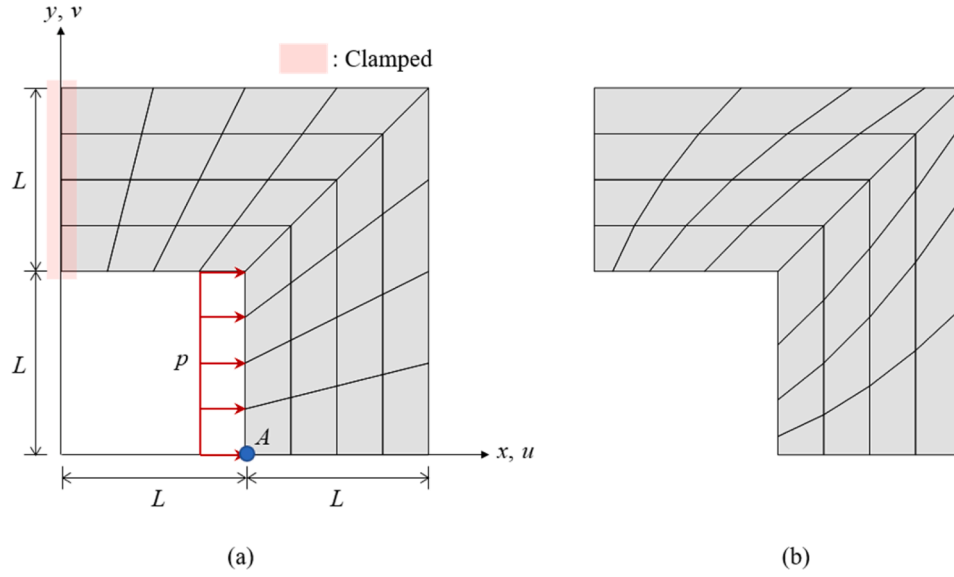


Fig. 10. L-shaped structure problem ($L = 50.0$, $E = 2.0 \times 10^{11}$, $\nu = 0.22$, and $p = 1.0$). (a) Regular mesh with $N = 4$, (b) Distorted mesh with $N = 4$.

with

$$a_A = \frac{\alpha(\alpha-1)}{2d}, a_B = \frac{\alpha(\alpha+1)}{2d}, a_C = \frac{\beta(\beta-1)}{2d}, a_D = \frac{\beta(\beta+1)}{2d}, a_E = \frac{2\alpha\beta}{d}, \quad (23)$$

$$\alpha = \frac{\mathbf{n} \cdot (\mathbf{x}_d \times \mathbf{x}_s)}{|\mathbf{x}_r \times \mathbf{x}_s|}, \beta = \frac{\mathbf{n} \cdot (\mathbf{x}_r \times \mathbf{x}_d)}{|\mathbf{x}_r \times \mathbf{x}_s|} \text{ and } d = \alpha^2 + \beta^2 - 1. \quad (24)$$

Using the assumed membrane strain field in Eqs. (20)–(22), the

assumed in-plane covariant strain is given by

$$\tilde{e}_{ij} = \tilde{e}_{ij}^m + t e_{ij}^{b1} + t^2 e_{ij}^{b2}. \quad (25)$$

Note that when the element geometry is flat, the assumed membrane strain $\tilde{e}_{ij}^m$ becomes identical to the displacement-based membrane strain e_{ij}^m .

The bending performance of the new MITC4+ shell element is significantly better than that of the MITC4 shell element when the

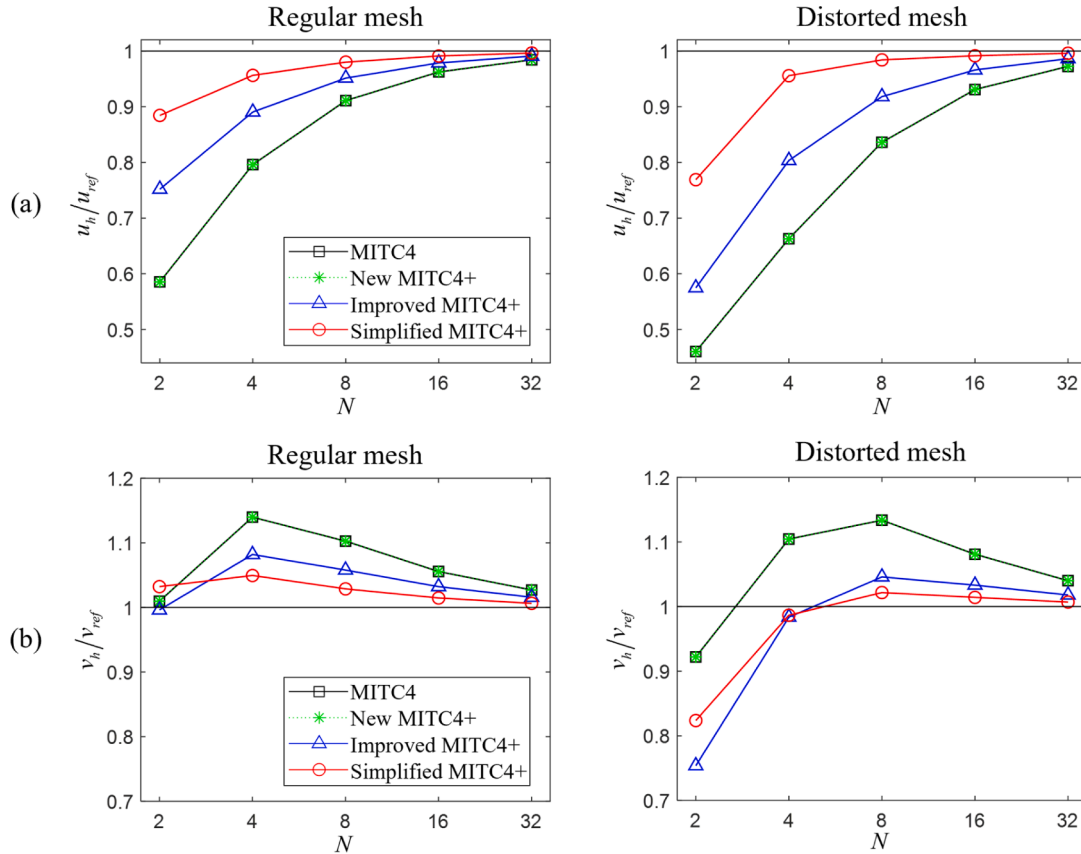


Fig. 11. Normalized (a) horizontal (u_h/u_{ref}) and (b) vertical (v_h/v_{ref}) displacements at point A in an L-shaped structure.

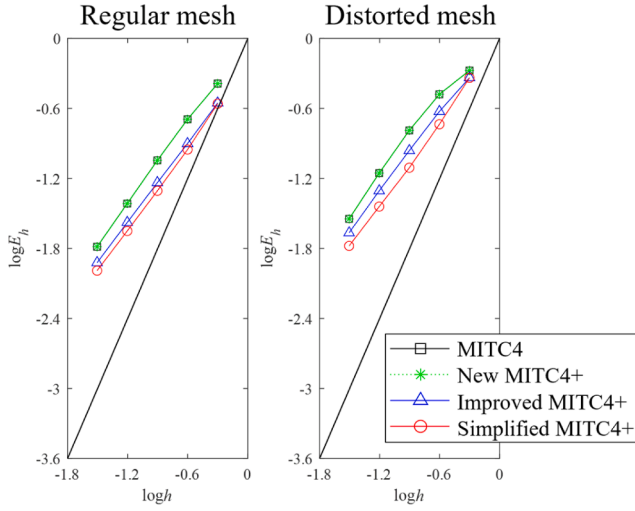


Fig. 12. Convergence curves for the L-shaped structure problem. The bold line denotes the optimal convergence rate.

geometry of the element is warped. The membrane behavior of the new MITC4+ shell element was improved by adopting the assumed strain field of the 2D MITC solid element, leading to the improved MITC4+ shell element [37]. In other words, in-plane shear locking is additionally treated in the shell element. However, this improvement complicates the formulation of the improved MITC4+ shell element. The assumed membrane strain field of the improved MITC4+ shell element is shown in Table 2 without detailed derivation.

3. Simplified MITC4+ shell element

In this section, the formulation of the simplified MITC4+ shell element is presented in detail and compared with that of the improved MITC4+ shell element. A simplified assumed membrane strain field is introduced and the geometry-dependent Gauss integration is adopted.

3.1. Simplified assumed membrane strains

In this study, a simple and effective assumed membrane strain field is proposed. A simplified assumed membrane strain field is constructed using the following strain components [37].

$$\bar{e}_{ij}^m = e_{kl}^m g_i^k g_j^l \text{ with } \hat{\mathbf{g}}^i \cdot \bar{\mathbf{g}}_j = \delta_j^i, \quad i, j, k, l = 1, 2, \quad (26)$$

$$\hat{\mathbf{g}}^i = \mathbf{g}^i(r, s, 0) \text{ and } \bar{\mathbf{g}}_i = \mathbf{g}_i(0, 0, 0), \quad (27)$$

where $\mathbf{g}^i$ is the contravariant base vector obtained using the relationship $\mathbf{g}^i \cdot \mathbf{g}_j = \delta_j^i$, $\hat{\mathbf{g}}^i$ is the contravariant base vector evaluated at the mid surface of the shell ($t = 0$), and $\bar{\mathbf{g}}_i$ is the covariant base vector evaluated at the element center ($r = s = t = 0$). In other words, the simplified assumed membrane strain is defined in the element-center covariant coordinate system.

First, the assumed membrane strain field of the 2D-MITC4 solid element is generalized with a variable k in the range $[0, 1]$ which determines the tying positions, as shown in Fig. 5(a).

$$\bar{e}_{rr}^{AS} = \bar{e}_{rr}^m(0, 0) + \lambda(r, s) \frac{\bar{e}_{rr}^m(0, k) - \bar{e}_{rr}^m(0, -k)}{2k} s, \quad (28)$$

$$\bar{e}_{ss}^{AS} = \bar{e}_{ss}^m(0, 0) + \lambda(r, s) \frac{\bar{e}_{ss}^m(k, 0) - \bar{e}_{ss}^m(-k, 0)}{2k} r, \quad (29)$$

$$\bar{e}_{rs}^{AS} = \bar{e}_{rs}^m(0, 0) \quad (30)$$

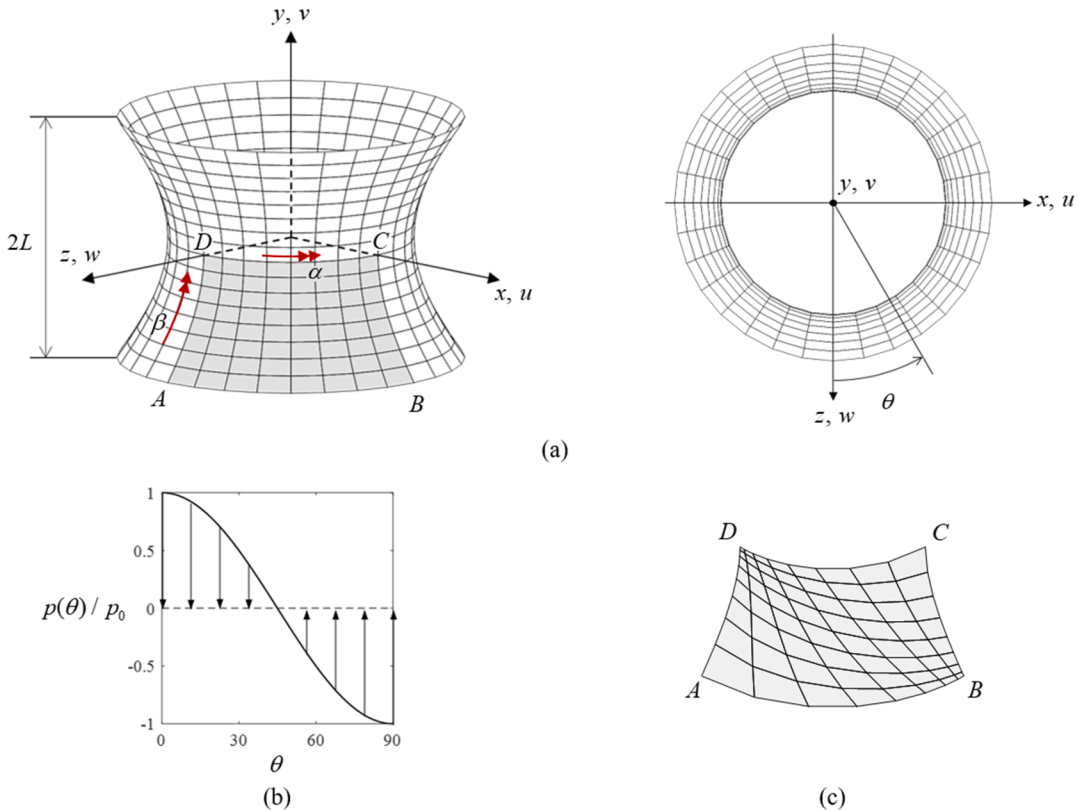


Fig. 13. Hyperboloid shell problem ($L = 1.0$, $E = 2.0 \times 10^{11}$, and $\nu = 1/3$). (a) Problem description and a 8×8 regular mesh, (b) Smoothly varying pressure loading ($p_0 = 1.0$), (c) An 8×8 distorted mesh.

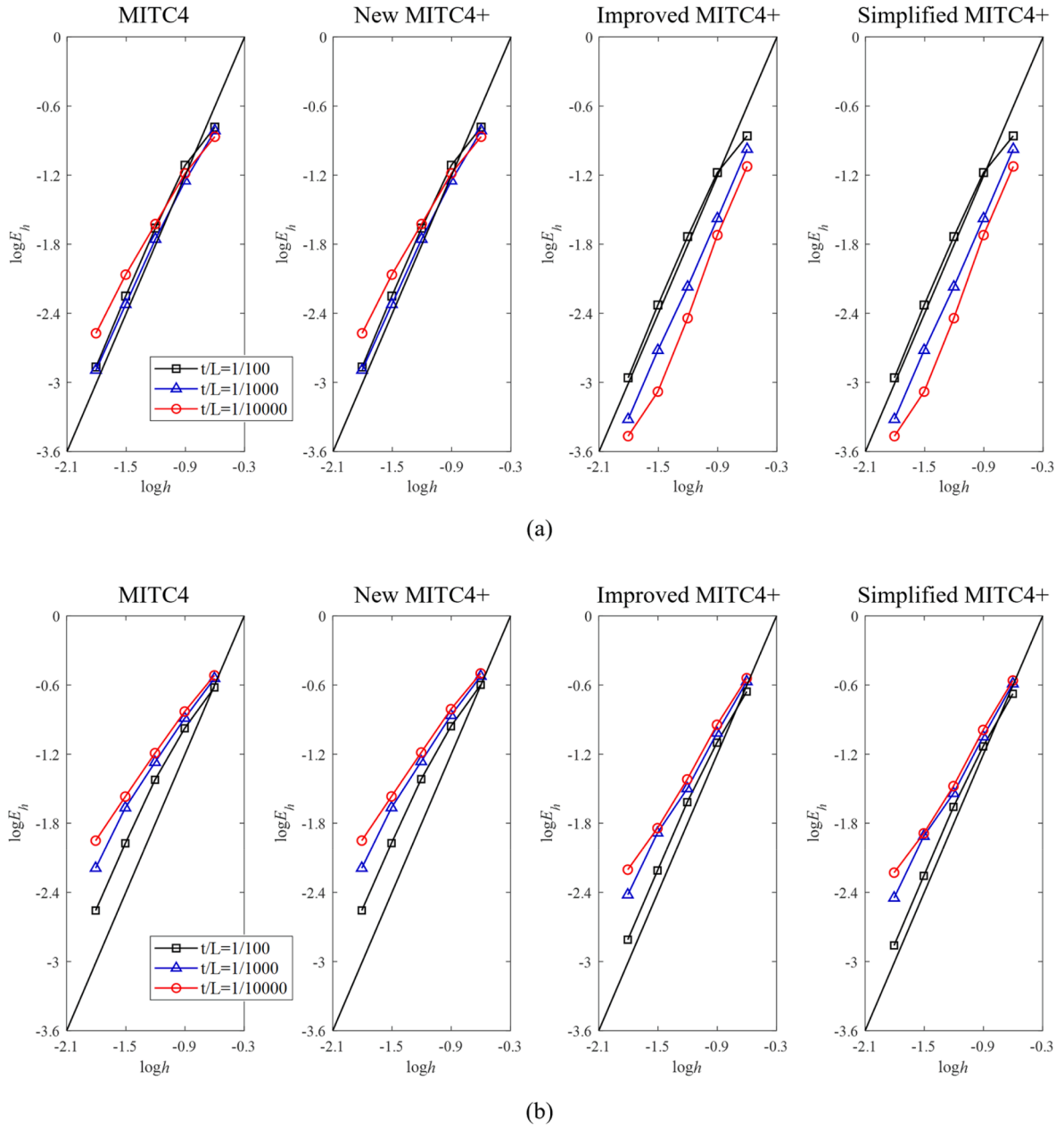


Fig. 14. Convergence curves for the hyperboloid shell problem with fully clamped condition considering (a) regular and (b) distorted meshes. The bold line denotes the optimal convergence rate.

$$\text{with } \lambda(r, s) = \frac{\det \mathbf{J}(0, 0)}{\det \mathbf{J}(r, s)} \text{ and } \mathbf{J}(r, s) = [\hat{\mathbf{g}}_r \quad \hat{\mathbf{g}}_s \quad \mathbf{n}], \quad (31)$$

where the superscript “AS” denotes the assumed membrane strains. Note that the assumed membrane strain field of the improved MITC4+ element can be obtained by setting k to $1/\sqrt{3}$, and the corresponding tying points (A)–(D) are shown in Fig. 5(b) [38].

By using the strain components calculated at the tying points, the assumed strain field in Eqs. (28)–(30) can be rewritten as follows:

$$\bar{e}_{rr}^{AS} = e_{rr}^{con} + \frac{1}{1 - k^2 \alpha^2} \lambda(r, s) (-2\alpha e_{rr}^{con} - 2\beta e_{rs}^{con} + e_{rr}^{lin}) s, \quad (32)$$

$$\bar{e}_{ss}^{AS} = e_{ss}^{con} + \frac{1}{1 - k^2 \beta^2} \lambda(r, s) (-2\beta e_{ss}^{con} - 2\alpha e_{rs}^{con} + e_{ss}^{lin}) r, \quad (33)$$

$$\bar{e}_{rs}^{AS} = e_{rs}^{con} \quad (34)$$

with the following strain coefficients [34],

$$e_{rr}^{con} = \mathbf{x}_r \cdot \mathbf{u}_r, \quad e_{ss}^{con} = \mathbf{x}_s \cdot \mathbf{u}_s, \quad e_{rs}^{con} = \frac{1}{2} (\mathbf{x}_r \cdot \mathbf{u}_s + \mathbf{x}_s \cdot \mathbf{u}_r), \quad (35)$$

$$e_{rr}^{lin} = \mathbf{x}_r \cdot \mathbf{u}_d + \mathbf{x}_d \cdot \mathbf{u}_r, \quad e_{ss}^{lin} = \mathbf{x}_s \cdot \mathbf{u}_d + \mathbf{x}_d \cdot \mathbf{u}_s. \quad (36)$$

The tying position affects only the second term on the right-hand side of Eqs. (32) and (33), which represents the bending behavior.

Setting $k = 0$ in Eqs. (32) and (33), the simplified assumed membrane strain field is obtained as

$$\bar{e}_{rr}^{AS} = e_{rr}^{con} + \lambda(r, s) (-2\alpha e_{rr}^{con} - 2\beta e_{rs}^{con} + e_{rr}^{lin}) s, \quad (37)$$

$$\bar{e}_{ss}^{AS} = e_{ss}^{con} + \lambda(r, s) (-2\beta e_{ss}^{con} - 2\alpha e_{rs}^{con} + e_{ss}^{lin}) r, \quad (38)$$

$$\bar{e}_{rs}^{AS} = e_{rs}^{con}. \quad (39)$$

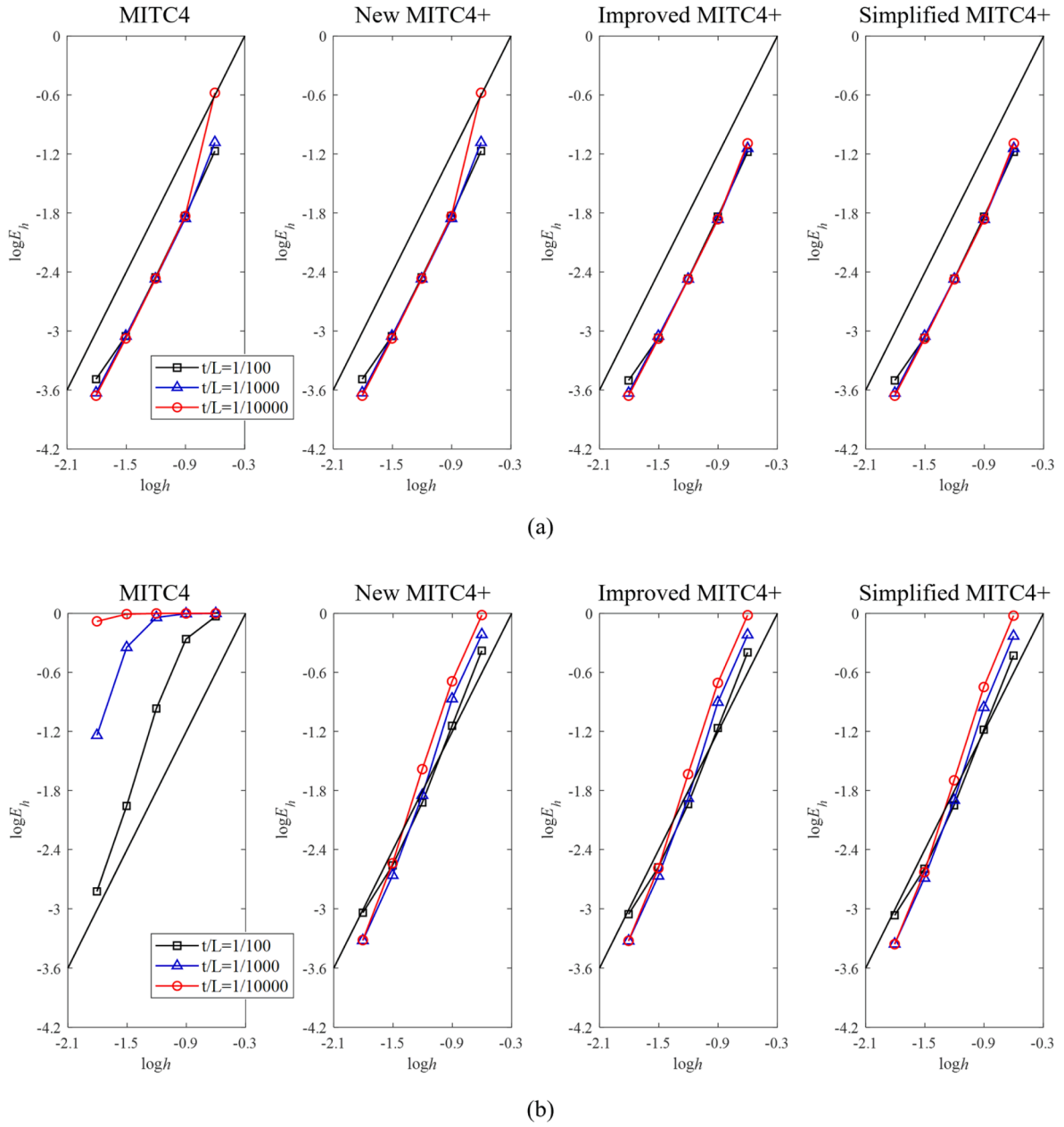


Fig. 15. Convergence curves for the hyperboloid shell problem with free boundary condition considering (a) regular and (b) distorted meshes. The bold line denotes the optimal convergence rate.

The simplified assumed membrane strain field is rewritten as

$$\begin{bmatrix} \bar{e}_{rr}^{AS} \\ \bar{e}_{ss}^{AS} \\ \bar{e}_{rs}^{AS} \end{bmatrix} = \mathbf{M}(r, s) \begin{bmatrix} e_{rr}^{con} & e_{ss}^{con} & e_{rs}^{con} & e_{rr}^{lin} & e_{ss}^{lin} \end{bmatrix}^T \quad (40)$$

with

$$\mathbf{M}(r, s) = \begin{bmatrix} 1 - 2\alpha\lambda s & 0 & -2\beta\lambda s & \lambda s & 0 \\ 0 & 1 - 2\beta\lambda r & -2\alpha\lambda r & 0 & \lambda r \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}, \quad (41)$$

where α and β are given by Eq. (24).

The relations between the strain coefficients and strains at tying points (A)–(E) are denoted as follows [34]:

$$e_{rr}^{con} + \bar{e}_{rs}^{bil} = \frac{1}{2}(e_{rr}^{(A)} + e_{rr}^{(B)}), \quad e_{rr}^{lin} = \frac{1}{2}(e_{rr}^{(A)} - e_{rr}^{(B)}), \quad (42)$$

$$e_{ss}^{con} + \bar{e}_{rs}^{bil} = \frac{1}{2}(e_{ss}^{(C)} + e_{ss}^{(D)}), \quad e_{ss}^{lin} = \frac{1}{2}(e_{ss}^{(C)} - e_{ss}^{(D)}), \quad (43)$$

$$e_{rs}^{con} = e_{rs}^{(E)} \quad (44)$$

with

$$\bar{e}_{rs}^{bil} = a_A e_{rr}^{(A)} + a_B e_{rr}^{(B)} + a_C e_{ss}^{(C)} + a_D e_{ss}^{(D)} + a_E e_{rs}^{(E)}, \quad (45)$$

where $\bar{e}_{rs}^{bil}$ is the assumed bilinear strain coefficient to alleviate membrane locking, e_{ij}^m are the membrane strain components in Eq. (9), and $a_A, a_B, \dots, a_E$ are given in Eq. (23).

Eqs. (42)–(44) are represented with a matrix $\mathbf{C}$

$$\begin{bmatrix} e_{rr}^{con} & e_{ss}^{con} & e_{rs}^{con} & e_{rr}^{lin} & e_{ss}^{lin} \end{bmatrix}^T = \mathbf{C} \begin{bmatrix} e_{rr}^{(A)} & e_{rr}^{(B)} & e_{ss}^{(C)} & e_{ss}^{(D)} & e_{rs}^{(E)} \end{bmatrix}^T \quad (46)$$

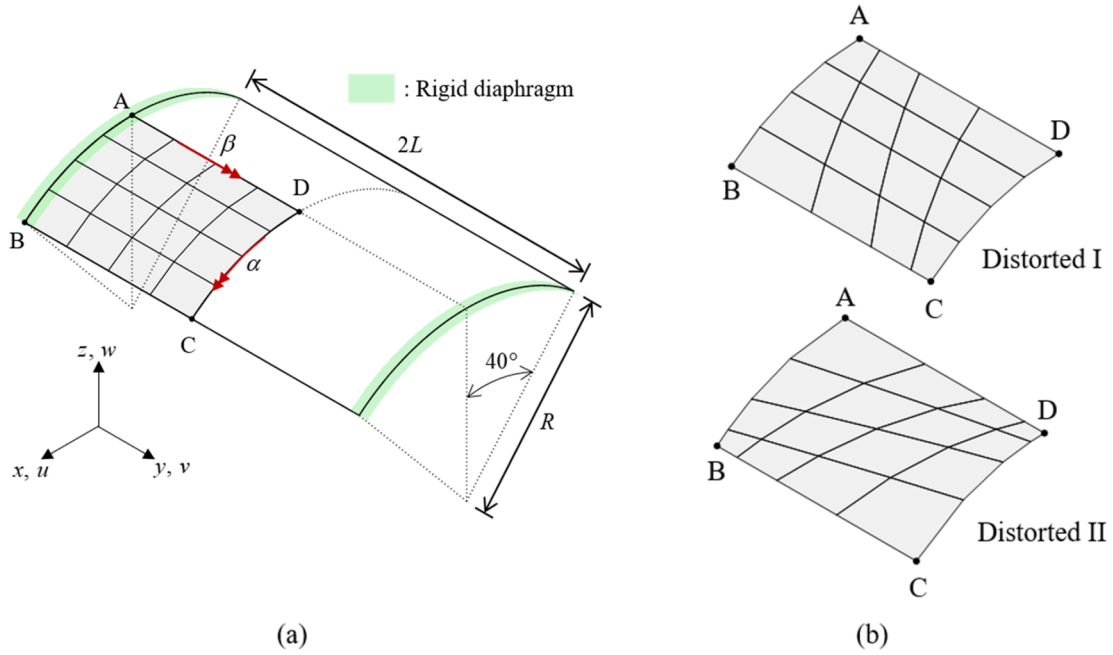


Fig. 16. Scordelis-Lo roof problem ($L = R = 25.0$, $E = 4.32 \times 10^8$, $\nu = 0.0$, and self-weight with $f_z = -90$ per unit area). (a) Problem description and a 4×4 regular mesh. (b) Two 4×4 distorted mesh patterns.

with

$$\mathbf{C} = \begin{bmatrix} 1/2 - a_A & 1/2 - a_B & -a_C & -a_D & -a_E \\ -a_A & -a_B & 1/2 - a_C & 1/2 - a_D & -a_E \\ 0 & 0 & 0 & 0 & 1 \\ 1/2 & -1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & -1/2 & 0 \end{bmatrix}. \quad (47)$$

To couple the new assumed membrane strains with other covariant strains, the membrane strains defined in the element-center covariant coordinate system are transformed as

$$e_{ij}^{AS} = \bar{e}_{kl}^{AS} (\mathbf{g}^k \cdot \mathbf{g}_i) (\mathbf{g}^l \cdot \mathbf{g}_j) \quad (48)$$

and the corresponding matrix form is

$$\begin{bmatrix} e_{rr}^{AS} \\ e_{ss}^{AS} \\ e_{rs}^{AS} \end{bmatrix} = \mathbf{Q} \begin{bmatrix} \bar{e}_{rr}^{AS} \\ \bar{e}_{ss}^{AS} \\ \bar{e}_{rs}^{AS} \end{bmatrix} \quad (49)$$

with

$$\mathbf{Q} = \begin{bmatrix} \alpha_s^2 & (\beta s)^2 & 2\beta\alpha_s s \\ (\alpha r)^2 & \beta_r^2 & 2\alpha\beta_r r \\ \alpha\alpha_{,r} & \beta\beta_{,s} & \alpha\beta_{rs} + \alpha_s\beta_r \end{bmatrix}, \quad \alpha_s = 1 + \alpha s, \quad \beta_r = 1 + \beta r. \quad (50)$$

The simplified assumed membrane strain field is expressed using the strain at the tying points (A)–(E) in Fig. 4 as follows:

$$\begin{bmatrix} e_{rr}^{AS} \\ e_{ss}^{AS} \\ e_{rs}^{AS} \end{bmatrix} = \mathbf{QMC} \begin{bmatrix} e_{rr}^m(A) & e_{rr}^m(B) & e_{ss}^m(C) & e_{ss}^m(D) & e_{rs}^m(E) \end{bmatrix}^T \quad (51)$$

and finally, the in-plane strain components are given by

$$\tilde{e}_{ij} = e_{ij}^{AS} + t e_{ij}^{b1} + t^2 e_{ij}^{b2}. \quad (52)$$

Table 2 lists the two matrices $\mathbf{M}$ and $\mathbf{C}$ used to construct the assumed membrane strain fields of the improved and simplified MITC4+ shell elements. Unlike the improved MITC4+ element, the simplified

MITC4+ element does not require the computation of ten coefficients, n_i and m_i in Table 2. As the strain field of the improved MITC4+ shell element is complicated, extending its formulation to nonlinear analysis is difficult. As additional information, the assumed membrane strains of the new MITC4+ element are provided in Appendix A.

3.2. Geometry-dependent Gauss integration

In general, the stiffness matrix of the 4-node quadrilateral shell element is obtained using 2×2 Gauss integration in the r - s plane and 2-point Gauss integration in the thickness direction [1–3]

$$\mathbf{K} = \sum_{i=1}^2 \sum_{j=1}^2 \sum_{k=1}^2 w_i w_j w_k \mathbf{F}(\xi_i, \xi_j, \xi_k) \quad (53)$$

$$\text{with } \xi_1 = \frac{1}{\sqrt{3}}, \quad \xi_2 = -\frac{1}{\sqrt{3}}, \quad w_1 = w_2 = 1$$

and

$$\mathbf{F}(r, s, t) = \mathbf{B}(r, s, t)^T \mathbf{DB}(r, s, t) \det \mathbf{J}(r, s, t), \quad (54)$$

where ξ_i and w_i denote Gauss points and the corresponding weights, respectively; $\mathbf{B}$ is the strain–displacement relation matrix; $\mathbf{J}$ is the Jacobian matrix; and $\mathbf{D}$ is the material law matrix for the shell elements [1].

Geometry-dependent Gauss integration is adopted to make the element less sensitive to geometry distortion [38]. For the shell element, the skewness of the element geometry and the corresponding adjustment parameter should be appropriately defined. Here, the skew angle θ of the element geometry is measured in the plane P defined by the characteristic geometry vectors $\mathbf{x}_r$ and $\mathbf{x}_s$, as shown in Fig. 6. The adjustment parameter μ is obtained as

$$\mu = \cos^2 \theta \quad \text{with } \cos \theta = \frac{1}{\|\mathbf{g}_r\| \|\mathbf{g}_s'\|} = \frac{1}{\|\mathbf{g}_s\| \|\mathbf{g}_r'\|} \quad (55)$$

With the adjustment parameter in Eq. (55), the numerical integration of the stiffness matrix of the simplified MITC4+ shell element is modified as follows:

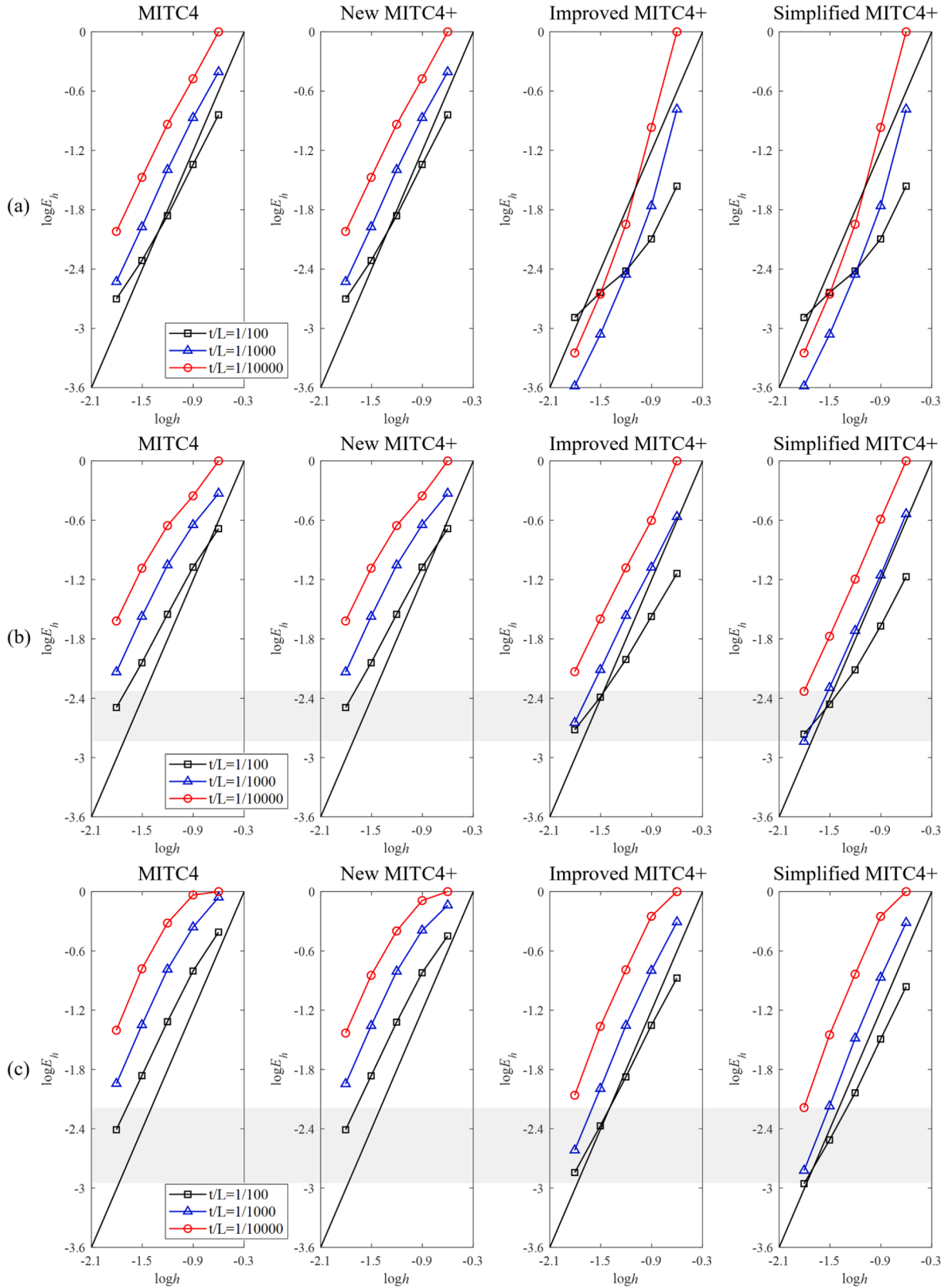


Fig. 17. Convergence curves for the Scordelis-Lo roof problem in (a) regular mesh, (b) distorted mesh I, and (c) distorted mesh II. The bold line denotes the optimal convergence rate.

$$\mathbf{K} = \sum_{k=1}^2 \sum_{j=1}^2 \sum_{i=1}^2 w_i w_j w_k \mathbf{F}(\xi_i', \xi_j', \xi_k') \quad (56)$$

with $\xi_1' = \mu \xi_1$, $\xi_2' = \mu \xi_2$, $w_1 = w_2 = 1$.

The integration positions did not change in the thickness direction, and the weight factors are the same. That is, geometry-dependent Gauss integration is adopted only for integration in the r -s plane.

To illustrate the effect of geometry-dependent Gauss integration on

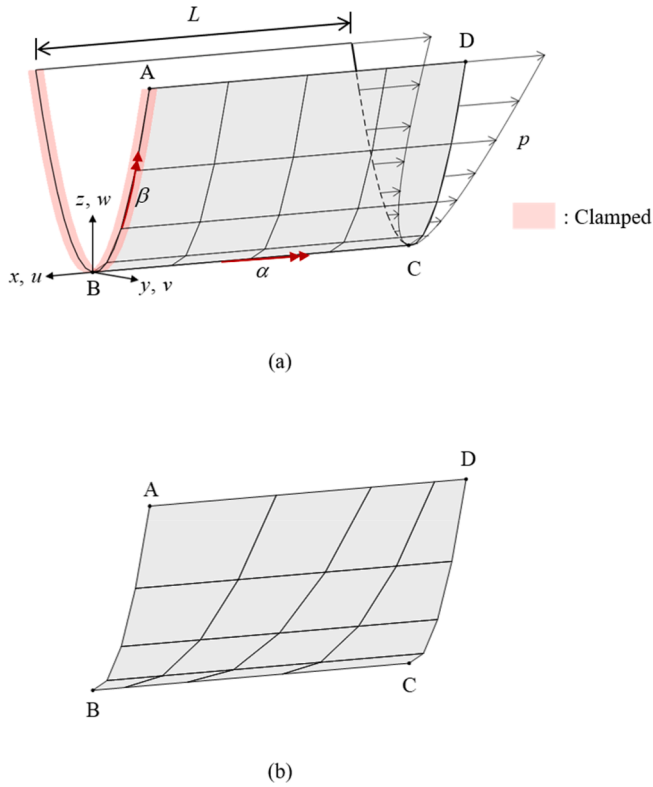


Fig. 18. Partly clamped parabolic cylinder problem ($L = 2$, $E = 2.0 \times 10^{11}$, and $\nu = 1/3$). (a) 4×4 regular mesh. (b) 4×4 distorted mesh.

the stiffness matrix, we conduct the eigen analysis of the stiffness matrix for a single simplified MITC4+ shell element depicted in Fig. 7(a). Young's modulus $E = 1.7472 \times 10^7$, Poisson's ratio $\nu = 0.3$, and thickness-to-length ratio $t/L = 10^{-4}$ are considered.

Table 3 shows the eigenvalues of the stiffness matrices of the simplified MITC4+ shell element according to the adjusting parameter. The eigenvalues are categorized by deformation modes corresponding to membrane, bending, transverse shearing, and in-plane twisting behaviors, see detailed deformed shapes in Ref. [12]. Fig. 7(b) illustrates the variation in normalized eigenvalues in relation to the adjusting parameter, where each eigenvalue is normalized using the corresponding eigenvalue with $\mu = 1$. As the adjusting parameter decreases, eigenvalues for the deformation modes BL1, BL2, TQ, BC4, ML1, and ML2 decrease, while those for other modes remain constant. This implies that the stiffness of only these six modes can be modulated by altering integration points.

For the nonlinear analysis of the simplified MITC4+ shell element, the adjustment parameter is obtained from the initial configuration because a total Lagrangian formulation is adopted in this study. If the updated Lagrangian formulation is adopted, the adjustment parameter can be updated.

When calculating the strain and stress in the element, the same adjustment parameter is used as follows:

$$\epsilon(r, s, t) = \mathbf{B}(\mu r, \mu s, t)\mathbf{U}, \quad (57)$$

$$\tau(r, s, t) = \mathbf{D}\mathbf{B}(\mu r, \mu s, t)\mathbf{U}, \quad (58)$$

where ϵ and τ denote the strain and stress vectors, respectively, and $\mathbf{U}$ denotes the nodal displacement vector of the element. Note that the strain at point (r, s, t) is evaluated at the adjusted point $(\mu r, \mu s, t)$, following the same method used for integrating the stiffness matrix in Eq. (56). Subsequently, the corresponding stress is calculated using the material law.

4. Basic numerical tests

Three basic numerical tests (patch, zero-energy mode, and isotropy) are conducted on the proposed MITC4+ shell element.

Membrane, bending, and shearing patch tests are performed using the geometry shown in Fig. 8 [1,32]. For each test, a minimum number of fixed boundary conditions is imposed to prevent rigid body motions, and nodal point loads are applied to produce constant stress fields [8]. The simplified MITC4+ shell element passes all the patch tests.

In the zero-energy mode test, a single shell element without any support should have only six zero-energy modes corresponding to the rigid body modes: three translations and three rotations [1–3]. The five elements in the mesh for the patch tests, shown in Fig. 8, are tested individually. The simplified MITC4+ shell element passes the zero-energy mode test.

All the finite elements should exhibit spatially isotropic behavior because the performance of an element should not be affected by the node numbering sequence [41]. Considering each element in the mesh shown in Fig. 8, the simplified MITC4+ shell element is tested and it passes the isotropy test.

5. Numerical examples

In this section, the performance of the simplified MITC4+ shell element is demonstrated by solving four linear problems (L-shaped structure, hyperboloid shell, Scordelis-Lo roof, and partly clamped parabolic cylinder) and three geometric nonlinear problems (thin curved beam, slit annular plate, and hemispherical shell).

The MITC4, new MITC4+, and improved MITC4+ shell elements are compared with the simplified MITC4+ shell element [4,34,37]. For each problem, both regular and distorted meshes are considered. The meshes are generated in the unit square domain as shown in Fig. 9, and then linearly mapped into the problem domain under consideration. In the regular patterns, each edge is divided into equal intervals with the length $h_x = h_y = 1/N$. In the distorted patterns, each edge is divided with the ratio of $L_1 : L_2 : L_3 : \dots : L_N = 1 : 2 : 3 : \dots : N$.

In this study, we did not develop the nonlinear analysis code for the improved MITC4+ shell element due to its intricate formulation, which precludes us from demonstrating its performance in nonlinear problems. Note that the nonlinear formulation of the improved MITC4+ shell element has not been studied yet.

When investigating convergence in linear problems, the s-norm proposed by Hiller and Bathe is employed to measure the error in the finite element solutions as follows [42]:

$$\|\mathbf{u}_{ref} - \mathbf{u}_h\|_s^2 = \int_{\Omega_{ref}} \Delta \epsilon^T \Delta \tau d\Omega_{ref} \quad (59)$$

$$\text{with } \Delta \epsilon = \epsilon_{ref} - \epsilon_h, \Delta \tau = \tau_{ref} - \tau_h,$$

where $\mathbf{u}_{ref}$ is the reference (displacement) solution, $\mathbf{u}_h$ is the solution obtained from finite element discretization, and Ω denotes the entire shell domain.

To compare the convergence rates of the shell elements, the normalized relative error is used

$$E_h = \frac{\|\mathbf{u}_{ref} - \mathbf{u}_h\|_s^2}{\|\mathbf{u}_{ref}\|_s^2}. \quad (60)$$

The theoretical convergence rate for the 4-node quadrilateral shell elements can be represented as

$$E_h \cong ch^2, \quad (61)$$

where h denotes the element size, and c is a constant independent of the element size. In this study, well-converged reference solutions are obtained using fine regular meshes of the MITC9 shell elements, which is known to satisfy the ellipticity and consistency conditions [43–48].

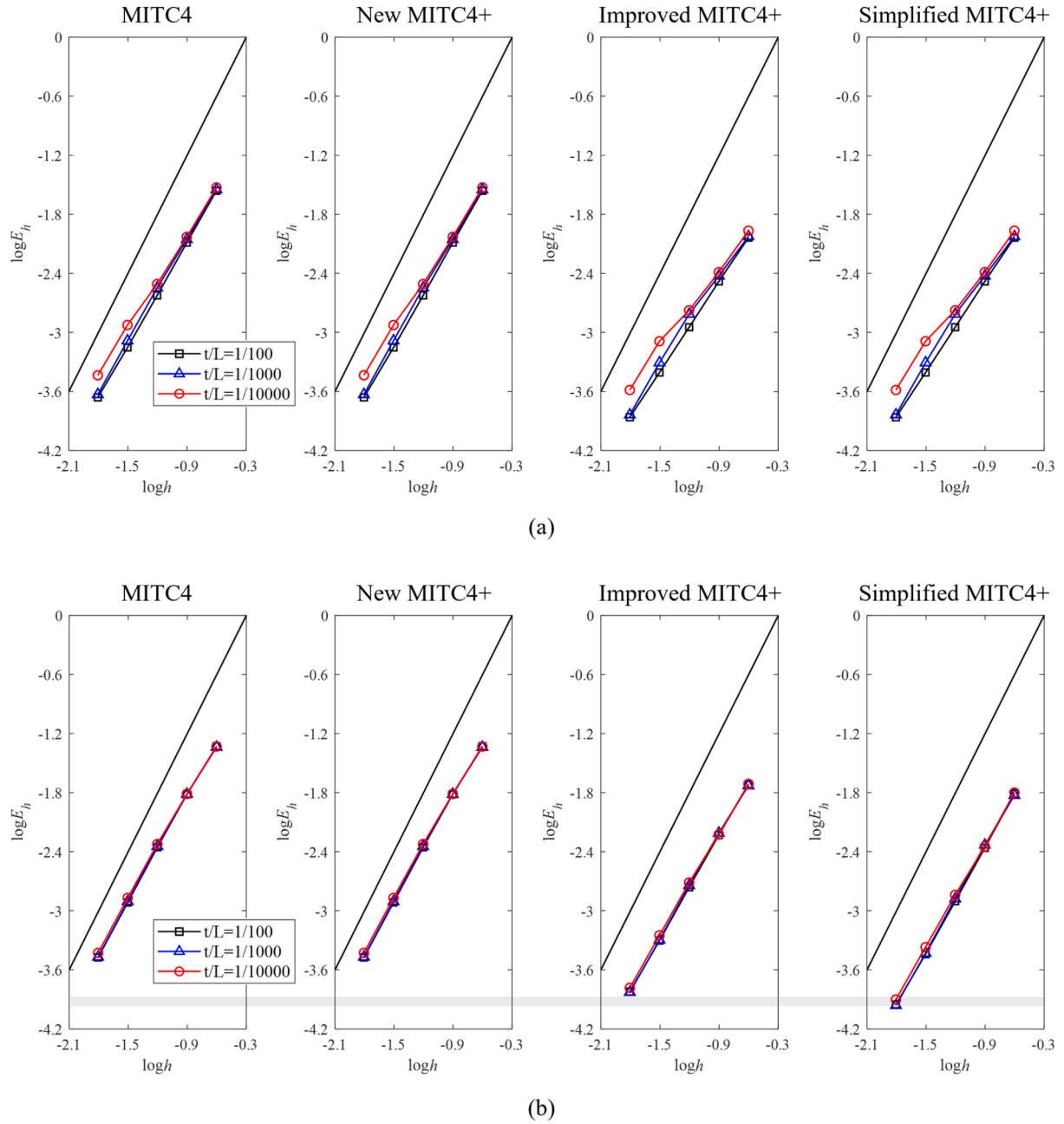


Fig. 19. Convergence curves for the partly clamped parabolic cylinder problem in (a) regular and (b) distorted meshes. The bold line denotes the optimal convergence rate.

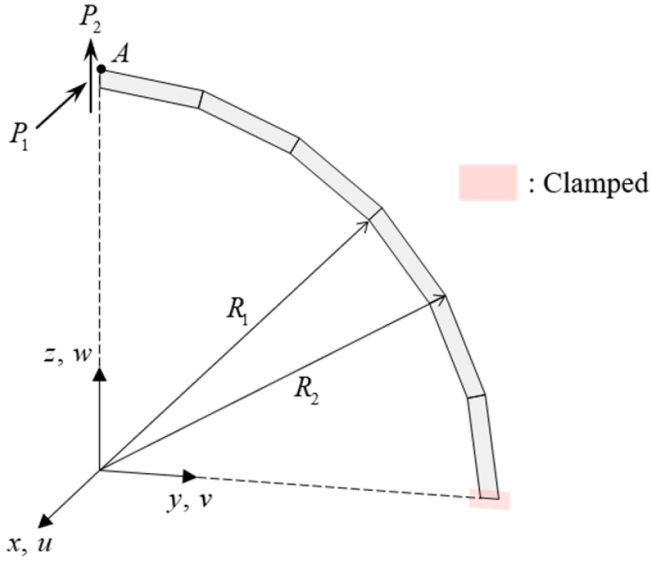


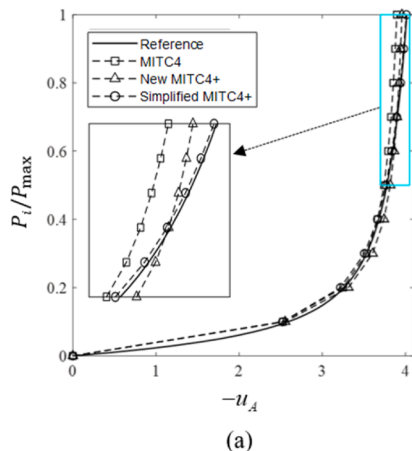
Fig. 20. Thin curved beam problem ($R_1 = 4.12$, $R_2 = 4.32$, $t = 0.1$, $E = 1.0 \times 10^7$ and $\nu = 0.25$) with a 1×6 regular mesh.

A total Lagrangian formulation is employed to analyze the large displacements and rotations. In addition, the Newton-Raphson method is used to solve the nonlinear equations with a tolerance of 0.1 percent in the relative incremental internal energy during each iteration.

5.1. L-shaped structure

As shown in Fig. 10, an L-shaped structure subjected to a distributed load $p = 1$ (force per length) is considered [8]. The left end is fixed. Young's modulus $E = 2.0 \times 10^{11}$ and Poisson's ratio $\nu = 0.22$ are provided. The structure is separated into two parts along the line $y = x$, and each part is modeled using $N \times N$ ($N = 2, 4, 8, 16$, and 32) element meshes. Both regular and distorted meshes are considered, as illustrated in Fig. 10.

The normalized horizontal and vertical displacements at point A are shown in Fig. 11. The convergence curves for the L-shaped structure problem with regular and distorted meshes are shown in Fig. 12. The reference solutions are obtained using a mesh of the MITC9 shell elements with $N = 96$. The simplified MITC4+ shell element provides more accurate solutions than the other shell elements in both regular and distorted meshes.



5.2. Hyperboloid shell

A hyperboloid shell problem is considered as shown in Fig. 13 [12,37]. The mid surface of the shell structure is given by

$$x^2 + z^2 = 1 + y^2, \quad y \in [-1, 1]. \quad (62)$$

The shell structure is subjected to a smoothly varying pressure $p(\theta) = p_0 \cos(2\theta)$, and the material properties are given by Young's modulus $E = 2.0 \times 10^{11}$ and Poisson's ratio $\nu = 1/3$. The hyperboloid shell exhibits two different asymptotic behaviors depending on the boundary conditions at both ends. This reveals the membrane-dominated or bending-dominated behavior under fully clamped or free boundary conditions, respectively.

Due to the symmetry of the problem, only one-eighth of the shell structure (ABCD) is modeled. The fully clamped boundary conditions are given as $u = \beta = 0$ along AD, $w = \beta = 0$ along BC, $v = \alpha = 0$ along CD, and $u = v = w = \alpha = \beta = 0$ along AB. The free boundary conditions are given as $u = \beta = 0$ along AD, $w = \beta = 0$ along BC, and $v = \alpha = 0$ along CD.

To investigate the convergence behavior using the s-norm, regular and distorted meshes are considered, as shown in Fig. 13(a) and (c), respectively. Three thickness-to-length ratios are considered: $t/L = 1/100$, $1/1000$, and $1/10000$. Finite element solutions are obtained using $N \times N$ element meshes with $N = 4, 8, 16, 32$, and 64 . The reference solutions are obtained using an 96×96 element mesh of the MITC9 shell elements. An element size of $h = L/N$ is used in the convergence curves.

Fig. 14 shows the convergence curves for the fully clamped hyperboloid shell with regular and distorted meshes. The convergence curves for free hyperboloid shell with regular and distorted meshes are shown in Fig. 15. Obtaining accurate solutions for the hyperboloid shell problem is difficult because of its doubly curved geometry with a negative Gaussian curvature.

All the considered elements provide accurate solutions for the clamped boundary condition, regardless of the mesh patterns. For the free boundary condition, the MITC4 element reveals an overly stiff behavior owing to membrane locking in distorted meshes. However, the new, improved, and simplified MITC4+ elements exhibit excellent convergence behavior in both the regular and distorted meshes. In particular, the improved and simplified MITC4+ shell elements exhibit similar convergence behaviors.

5.3. Scordelis-Lo roof

The Scordelis-Lo roof problem, shown in Fig. 16, is considered [36,46]. The structure is part of a cylinder with length $L = 25.0$, radius $R = 25.0$, and uniform thickness t . The roof is subjected to a self-weight

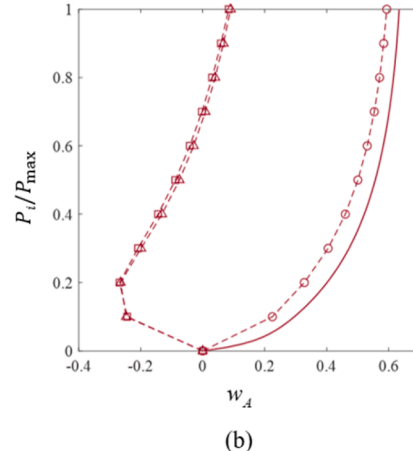


Fig. 21. Load-displacement curves for the thin curved beam problem. (a) $-u_A$ and (b) w_A .

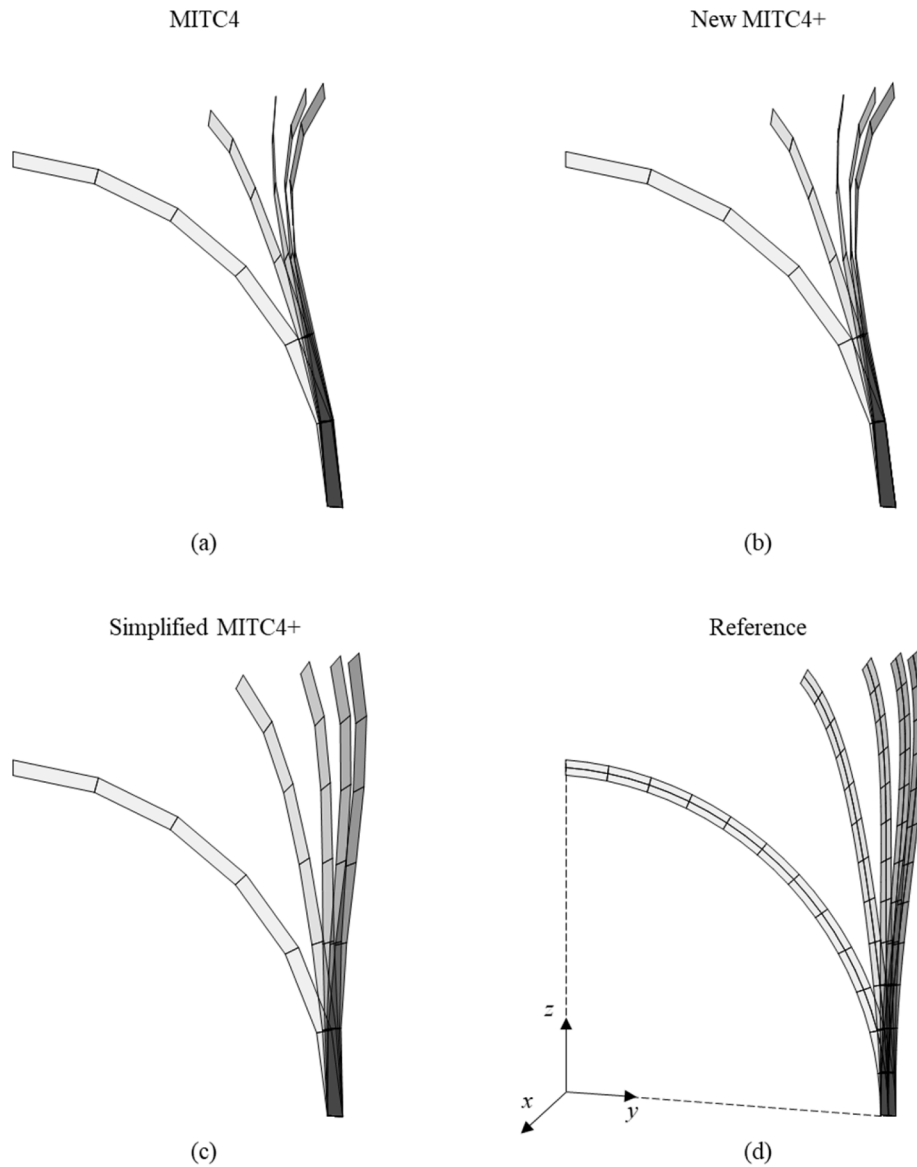


Fig. 22. Deformed shapes of the thin curved beam at several load steps obtained using (a) MITC4, (b) new MITC4+, and (c) simplified MITC4+ shell elements. (d) Reference shapes obtained using the MITC9 shell element.

loading $f_z = -90$ per unit area. Young's modulus $E = 4.32 \times 10^8$ and Poisson's ratio $\nu = 0.0$ are used. Three thickness-to-length ratios ($t/L = 1/100, 1/1000$, and $1/10000$) are considered.

Due to symmetry, only one-quarter of the structure corresponding to the region ABCD in Fig. 16(a) is modeled for analysis. For the rigid diaphragm boundary conditions, $u = \beta = 0$ along AD, $v = \alpha = 0$ along CD, and $u = w = 0$ along AB are considered. The regular and distorted mesh patterns shown in Fig. 16 are considered. For all mesh patterns, $N \times N$ element meshes with $N = 4, 8, 16, 32$, and 64 are used. The reference solutions are calculated using a 96×96 element mesh of the MITC9 shell elements. The Scordelis-Lo roof problem is known to show bending and membrane mixed behavior [46].

Fig. 17 shows the convergence curves for the three mesh patterns: regular mesh, distorted mesh I, and distorted mesh II. For the regular mesh case, the improved and simplified MITC4+ shell elements provide more accurate solutions. Note that the performances of the two elements are almost equal. However, when considering the distorted meshes I and II, the simplified MITC4+ shell element provides slightly more accurate solutions than the improved MITC4+ shell element.

5.4. Partly clamped parabolic cylinder

A parabolic cylinder of length $L = 2$ is considered, as shown in Fig. 18. The mid surface of the structure is given by

$$z = 4y^2, \quad y \in [-0.5, 0.5]. \quad (63)$$

The structure is clamped along the edge at $x = 0$ and subjected to a distributed line force $\mathbf{p}_x = -p_0 z \mathbf{e}_x$ with $p_0 = 1.0$ (force per length) along the other edge at $x = -L$.

Young's modulus and Poisson's ratio are $E = 2.0 \times 10^{11}$ and $\nu = 1/3$, respectively. Three different thickness-to-length ratios are considered: $t/L = 1/100, 1/1000$, and $1/10000$. Regular and distorted meshes are employed as shown in Fig. 18(a) and (b), respectively. Due to symmetry, only half of the problem domain is modeled. The following boundary conditions are considered: $v = \alpha = 0$ along BC and $u = v = w = \alpha = \beta = 0$ along AB.

Fig. 19(a) and (b) show the convergence curves for the partly clamped parabolic cylinder problem considering regular and distorted meshes, respectively. The $N \times N$ element meshes are used with $N = 4, 8, 16, 32$, and 64 . The reference solutions are obtained using the 96×96

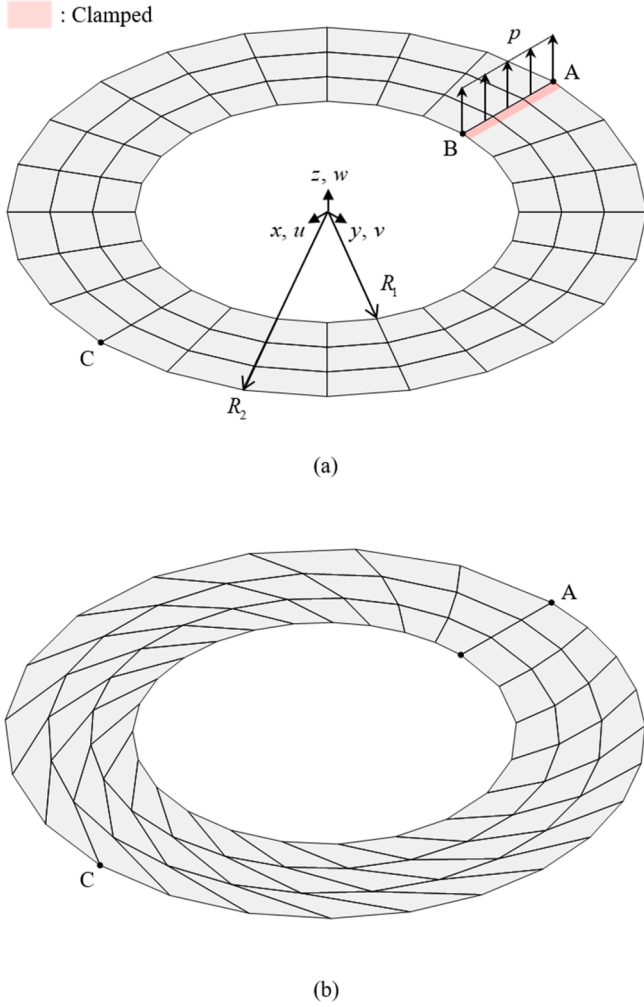
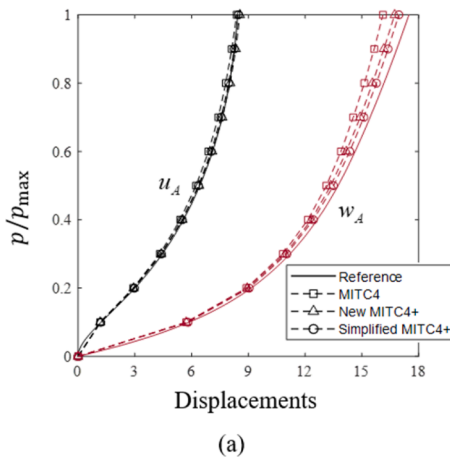


Fig. 23. Slit annular plate problem ($R_1 = 6$, $R_2 = 10$, $t = 0.03$, $E = 2.1 \times 10^7$, $\nu = 0$ and $p = 0.8$). (a) Problem descriptions and 3×24 regular mesh. (b) 3×24 distorted mesh.



element mesh of the MITC9 shell elements. An element size of $h = 1/N$ is used in the convergence curves. Although all the considered elements provide accurate solutions, the simplified MITC4+ shell element provides better solutions than the others when distorted meshes are considered.

5.5. Thin curved beam

A thin curved beam shown in Fig. 20 is considered [49,50]. The beam is clamped at one end and subjected to two forces, P_1 and P_2 , in the x - and z - directions, respectively. The material properties are given by Young's modulus $E = 1.0 \times 10^7$ and Poisson's ratio $\nu = 0.25$. The thickness of the beam is set as $t = 0.1$. The two forces increase simultaneously from $P_1 = P_2 = 0$ to 100.

The curved beam is modeled using a 1×6 regular mesh. Geometrical nonlinear analyses are performed using MITC4, new MITC4+, and simplified MITC4+ shell elements. The reference solution is obtained using a 2×12 regular mesh of the MITC9 shell elements.

Fig. 21 shows the load-displacement curves at point A for the thin curved beam. In addition, Fig. 22 shows the deformed shapes at five load steps: $P_1 = P_2 = 0$, $0.25P_{\max}$, $0.5P_{\max}$, $0.75P_{\max}$, and $P_{\max}$ with $P_{\max} = 100$. Unlike the MITC4 and new MITC4+ shell elements, the deformed shapes obtained using the simplified MITC4+ element agree well with the reference shapes.

5.6. Slit annular plate

As shown in Fig. 23, an annular plate with a slit along AB is considered. A vertical distributed load $p = 0.8$ (force per length) is applied to one side of the slit, and the other side of the slit is clamped [25,39,51]. Young's modulus $E = 2.1 \times 10^7$ and Poisson's ratio $\nu = 0$ are used. The thickness of the structure is $t = 0.03$.

Through geometrical nonlinear analyses, finite element solutions are obtained using the MITC4, new MITC4+, and simplified MITC4+ shell elements, considering both 3×24 regular and distorted meshes. The distorted mesh pattern is shown in Fig. 23(b).

Figs. 24 and 25 show the load-displacement curves at points A and C, respectively, for both regular and distorted meshes. The reference solutions are obtained using a 12×60 regular mesh of the MITC9 shell

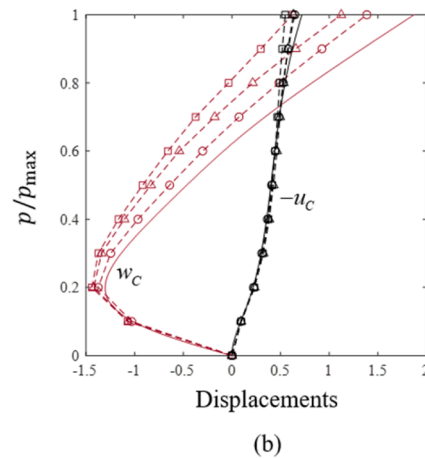


Fig. 24. Load-displacement curves for the slit annular plate problem using the 3×24 regular mesh. (a) Displacements at point A. (b) Displacements at point C.

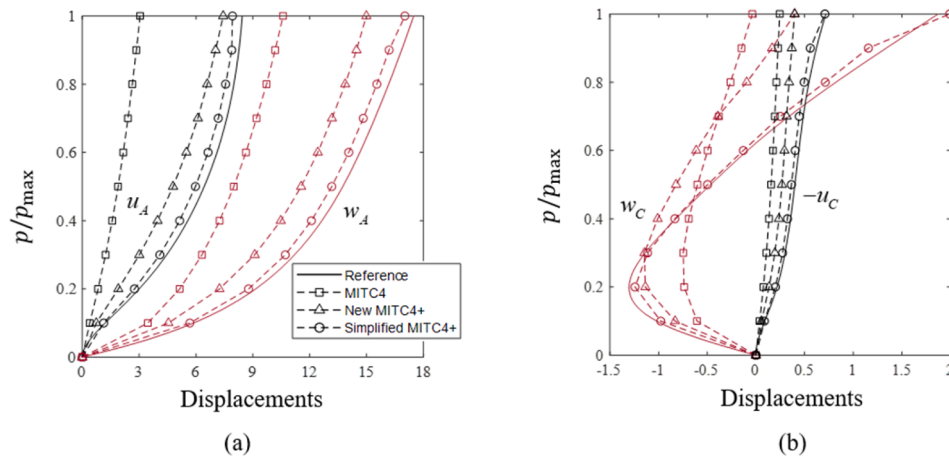


Fig. 25. Load-displacement curves for the slit annular plate problem using the 3×24 distorted mesh. (a) Displacements at point A. (b) Displacements at point C.

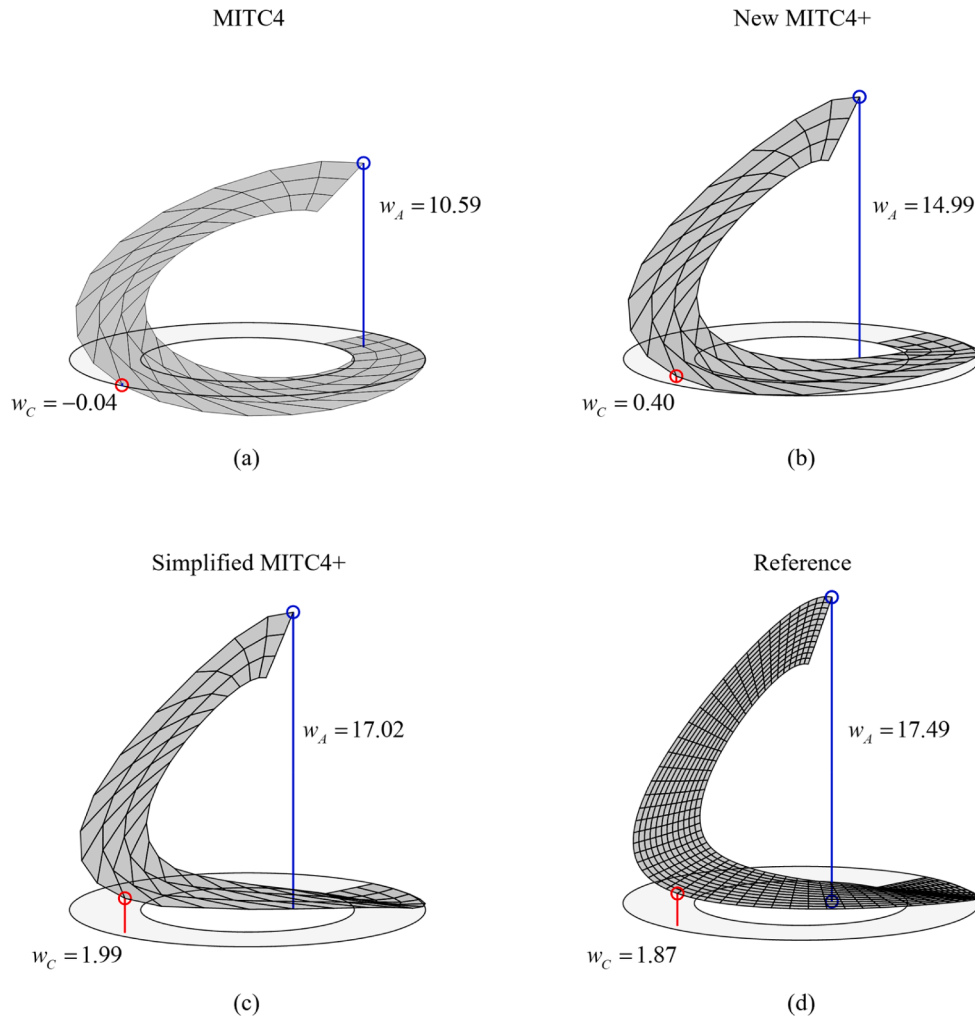


Fig. 26. Deformed shapes of the slit annular plate calculated using the 3×24 distorted mesh of (a) MITC4, (b) new MITC4+, (c) simplified MITC4+ elements, and (d) Reference solution obtained using the MITC9 shell element.

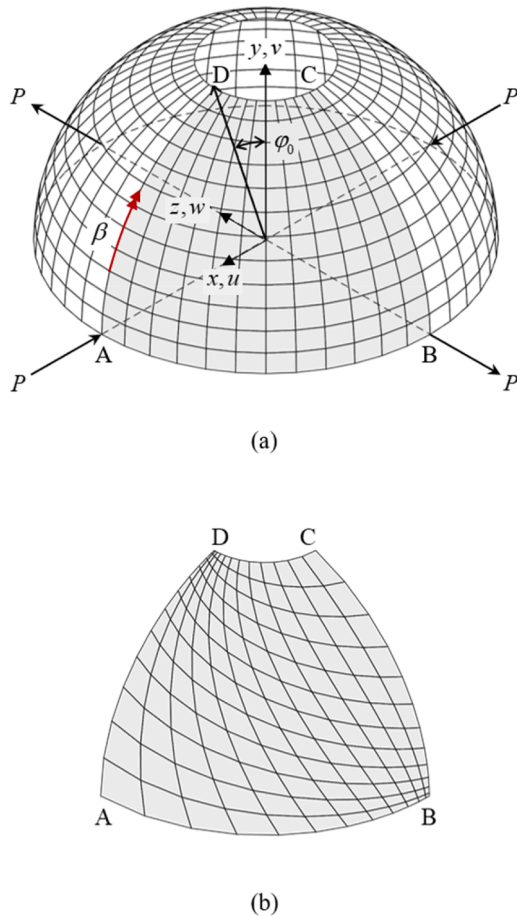


Fig. 27. Hemispherical shell problem ($R = 10$, $\varphi_0 = 18^\circ$, $t = 0.04$, $E = 6.825 \times 10^7$, $\nu = 0.3$, $P = 400$). (a) Problem description and 12×12 regular mesh. (b) 12×12 distorted mesh.

elements. Overall, the simplified MITC4+ shell element exhibits better performance than the other elements. In particular, the improvement in the predictive capability is significant in the distorted mesh case. The final deformed shapes calculated for the distorted mesh are shown in Fig. 26.

5.7. Hemispherical shell

The geometry of the hemispherical shell is shown in Fig. 27 [25,52]. The hemispherical shell has an 18° hole at its top and is subjected to two pairs of opposite radial concentrated loads of $P = 400$. This problem has been used extensively to investigate whether an element can represent rigid body rotation and inextensible bending modes. Young's modulus $E = 6.825 \times 10^7$ and Poisson's ratio $\nu = 0.3$ are used. Both regular and distorted meshes are considered, as illustrated in Fig. 27.

Due to symmetry, only one-quarter of a 12×12 element mesh is employed, which is the region ABCD in Fig. 27. The boundary conditions are given by $w = \beta = 0$ along AD, $u = \beta = 0$ along BC, and $v = 0$ at point A. Geometrical nonlinear analyses are performed using the MITC4, new MITC4+, and simplified MITC4+ shell elements. The reference solutions are obtained using a 36×36 regular mesh of the MITC9 shell elements.

In Fig. 28, the load-displacement curves at points A and B are

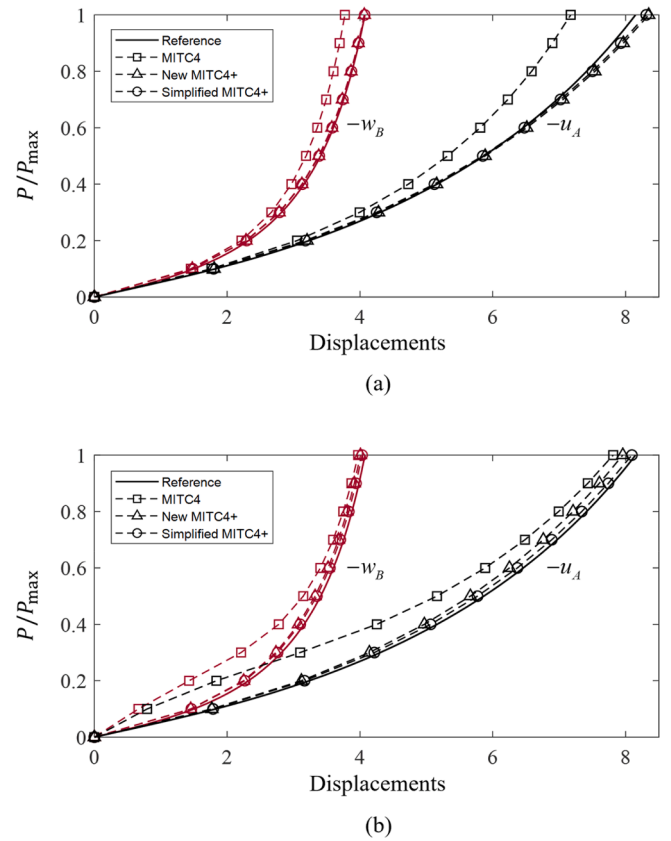


Fig. 28. Load-displacement curves at points A and B for the hemispherical shell problem calculated using (a) regular and (b) distorted meshes.

compared with reference solutions. Fig. 29 shows the deformed shapes of the entire structure calculated using the simplified MITC4+ shell element at several load steps. The new and simplified MITC4+ shell elements provide more accurate solutions than the MITC4 shell element.

6. Concluding remarks

In this study, a simplified MITC4+ shell element was developed for both linear and nonlinear analyses. A simplified assumed strain field was derived by merging the tying points into the element center. Moreover, a geometry-dependent Gauss integration scheme was adopted to enhance the membrane behavior, particularly in distorted meshes. The proposed simplified MITC4+ shell element successfully passed all the basic tests, including isotropy, patch, and zero-energy mode tests. In regular meshes, the convergence behavior of the simplified MITC4+ shell element is similar to that of the improved MITC4+ shell element. The simplified MITC4+ shell element outperformed the improved MITC4+ shell element in distorted meshes. Unlike the improved MITC4+ shell element, the formulation of the simplified MITC4+ shell element can be easily extended to nonlinear analysis. Its performance in nonlinear analysis was excellent.

It is valuable to employ fundamental ideas for the simplified MITC4+ shell element to improve the formulation of a 3D MITC solid element [36]. Moreover, studying the mathematical insights of various MITC4+ shell element formulations is an important research matter.

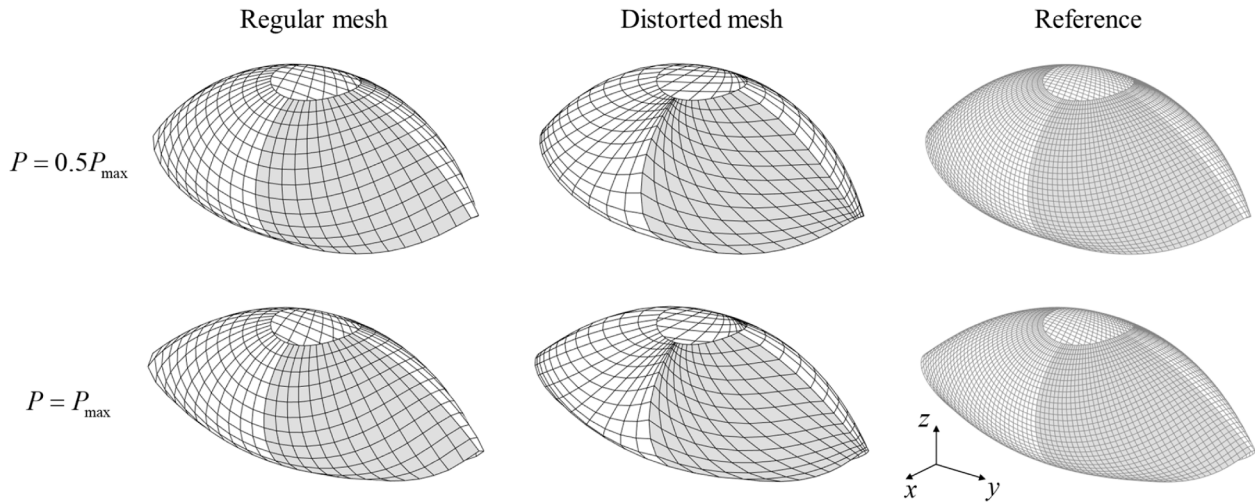


Fig. 29. Deformed shapes of the hemispherical shell calculated using the 12×12 regular and distorted meshes of the simplified MITC4+ shell element at load steps $P = 0.5P_{\max}$ and $P = P_{\max}$.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgments

This work was supported by the Korea Institute for Advancement of Technology (KIAT) grant funded by the Korean Government (MOTIE) (P0017006, The Competency Development Program for Industry Specialist) and by the National Research Foundation of Korea (NRF) grant funded by the Korean Government (MSIT) (No. NRF2018R1A2B3005328).

Appendix A. Assumed membrane strains of the new MITC4+ shell element

The assumed membrane strains of the new MITC4+ shell element [34] are given by the following equation:

$$\begin{bmatrix} \tilde{e}_{rr}^m \\ \tilde{e}_{ss}^m \\ \tilde{e}_{rs}^m \end{bmatrix} = \mathbf{MC} \begin{bmatrix} e_{rr}^{m(A)} & e_{rr}^{m(B)} & e_{ss}^{m(C)} & e_{ss}^{m(D)} & e_{rs}^{m(E)} \end{bmatrix}^T \quad (\text{A1})$$

with

$$\mathbf{M} = \begin{bmatrix} 1 & 0 & 0 & s & 0 & s^2 \\ 0 & 1 & 0 & 0 & r & r^2 \\ 0 & 0 & 1 & r/2 & s/2 & rs \end{bmatrix}, \quad (\text{A2})$$

$$\mathbf{C} = \begin{bmatrix} 1/2 - a_A & 1/2 - a_B & -a_C & -a_D & -a_E \\ -a_A & -a_B & 1/2 - a_C & 1/2 - a_D & -a_E \\ 0 & 0 & 0 & 0 & 1 \\ 1/2 & -1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & -1/2 & 0 \\ a_A & a_B & a_C & a_D & a_E \end{bmatrix}. \quad (\text{A3})$$

References

- [1] Bathe KJ. Finite element procedures. Prentice Hall; 1996, 2nd ed. K.J. Bathe, Watertown, MA; 2014 and Higher Education Press, China; 2016.
- [2] Hughes T.J.R. The finite element method: linear static and dynamic finite element analysis. Mineola, NY: Dover Publications; 2000.
- [3] Cook RD. Concepts and applications of finite element analysis. New York: John Wiley & Sons; 2007.
- [4] Dvorkin EN, Bathe KJ. A continuum mechanics based four-node shell element for general nonlinear analysis. Eng Comput 1984;1:77–88.
- [5] Lee PS, Bathe KJ. Development of MITC isotropic triangular shell finite elements. Comput Struct 2004;82:945–62.
- [6] Bucleml ML, Bathe KJ. Finite element analysis of shell structures. Arch Comput Meth Eng 1997;4:3–61.
- [7] Dvorkin EN. Nonlinear analysis of shells using the MITC formulation. Arch Comput Meth Eng 1995;2:1–50.
- [8] Lee C, Kim S, Lee PS. The strain-smoothed 4-node quadrilateral finite element. Comput Methods Appl Mech Eng 2021;373:113481.
- [9] Ko Y, Lee Y, Lee PS, Bathe KJ. Performance of the MITC3+ and MITC4+ shell elements in widely-used benchmark problems. Comput Struct 2017;193:187–206.
- [10] Ko Y, Lee PS, Bathe KJ. The MITC4+ shell element and its performance. Comput Struct 2016;169:57–68.
- [11] Mi Y, Yu X. Isogeometric MITC shell. Comput Methods Appl Mech Eng 2021;377: 113693.

- [12] Lee Y, Jeon HM, Lee PS, Bathe KJ. The modal behavior of the MITC3+ triangular shell element. *Comput Struct* 2015;153:148–64.
- [13] Lee PS, Bathe KJ. Insight into finite element shell discretizations by use of the “basic shell mathematical model”. *Comput Struct* 2005;83:69–90.
- [14] Zienkiewicz OC, Taylor RL, Too JM. Reduced integration technique in general analysis of plates and shells. *Int J Numer Meth Eng* 1971;3:275–90.
- [15] Belytschko T, Tsay C-S-A. stabilization procedure for the quadrilateral plate element with one-point quadrature. *Int J Numer Meth Eng* 1983;19:405–19.
- [16] Reddy BD, Küssner M. The four-noded quadrilateral with a 2×1 integration rule: Application to plates and other problems. *Comput Methods Appl Mech Eng* 1997;149:101–12.
- [17] Malkus DS, Hughes TJR. Mixed finite element methods - Reduced and selective integration techniques: A unification of concepts. *Comput Methods Appl Mech Eng* 1978;15:63–81.
- [18] Belytschko T, Leviathan I. Projection schemes for one-point quadrature shell elements. *Comput Methods Appl Mech Eng* 1994;115:277–86.
- [19] Belytschko T, Leviathan I. Physical stabilization of the 4-node shell element with one point quadrature. *Comput Methods Appl Mech Eng* 1994;113:321–50.
- [20] Roh HY, Cho M. The application of geometrically exact shell elements to B-spline surfaces. *Comput Methods Appl Mech Eng* 2004;193:2261–99.
- [21] Choi CK, Paik JG. An efficient four node degenerated shell element based on the assumed covariant strain. *Struct Eng Mech* 1994;2:17–34.
- [22] Kulikov GM, Plotnikova SV. A family of ANS four-node exact geometry shell elements in general convected curvilinear coordinates. *Int J Numer Meth Eng* 2010;83:1376–406.
- [23] Belytschko T, Wong BL, Stolarski H. Assumed strain stabilization procedure for the 9-node Lagrange shell element. *Int J Numer Meth Eng* 1989;28:385–414.
- [24] Prathap G. A C0 continuous four-noded cylindrical shell element. *Comput Struct* 1985;21:995–9.
- [25] Wagner W, Gruttmann F. An improved quadrilateral shell element based on the Hu–Washizu functional. *Adv Model Simul Eng Sci* 2020;7.
- [26] Bucalem ML, Bathe K-J. Higher-order MITC general shell elements. *Int J Numer Meth Eng* 1993;36:3729–54.
- [27] Beirão da Veiga L, Chapelle D, Paris SI. Towards improving the MITC6 triangular shell element. *Comput Struct* 2007;85:1589–610.
- [28] Lee Y, Lee PS, Bathe KJ. The MITC3+ shell element and its performance. *Comput Struct* 2014;138:12–23.
- [29] Lee C, Lee PS. The strain-smoothed MITC3+shell finite element. *Comput Struct* 2019;223:106096.
- [30] Lee C, Lee DH, Lee PS. The strain-smoothed MITC3+ shell element in nonlinear analysis. *Comput Struct* 2022;266:106768.
- [31] Bathe KJ, Dvorkin EN. A four-node plate bending element based on Mindlin/Reissner plate theory and a mixed interpolation. *Int J Numer Meth Eng* 1985;21:367–83.
- [32] Jeon HM, Lee PS, Bathe KJ. The MITC3 shell finite element enriched by interpolation covers. *Comput Struct* 2014;134:128–42.
- [33] Bathe KJ, Dvorkin EN. A formulation of general shell elements—the use of mixed interpolation of tensorial components. *Int J Numer Meth Eng* 1986;22:697–722.
- [34] Ko Y, Lee PS, Bathe KJ. A new MITC4+ shell element. *Comput Struct* 2017;182:404–18.
- [35] Dvorkin EN, Vassoloã SI. A quadrilateral 2-D finite element based on mixed interpolation of tensorial components. *Eng Comput* 1989;6:217–24.
- [36] Ko Y, Bathe KJ. A new 8-node element for analysis of three-dimensional solids. *Comput Struct* 2018;202:85–104.
- [37] Ko Y, Lee PS, Bathe KJ. A new 4-node MITC element for analysis of two-dimensional solids and its formulation in a shell element. *Comput Struct* 2017;192:34–49.
- [38] Choi HG, Lee PS. Towards improving the 2D-MITC4 element for analysis of plane stress and strain problems. *Comput Struct* 2023;275:106933.
- [39] Jeon HM, Lee Y, Lee PS, Bathe KJ. The MITC3+ shell element in geometric nonlinear analysis. *Comput Struct* 2015;146:91–104.
- [40] Ko Y, Lee PS, Bathe KJ. The MITC4+ shell element in geometric nonlinear analysis. *Comput Struct* 2017;185:1–14.
- [41] Lee C, Lee PS. A new strain smoothing method for triangular and tetrahedral finite elements. *Comput Methods Appl Mech Eng* 2018;341:939–55.
- [42] Hiller JF, Bathe KJ. Measuring convergence of mixed finite element discretizations: An application to shell structures. *Comput Struct* 2003;81:639–54.
- [43] Bathe KJ, Lee PS. Measuring the convergence behavior of shell analysis schemes. *Comput Struct* 2011;89:285–301.
- [44] Bathe KJ, Iosilevich A, Chapelle D. Inf-sup test for shell finite elements. *Comput Struct* 2000;75:439–56.
- [45] Bathe KJ. The inf-sup condition and its evaluation for mixed finite element methods. *Comput Struct* 2001;79:243–52.
- [46] Lee PS, Bathe KJ. On the asymptotic behavior of shell structures and the evaluation in finite element solutions. *Comput Struct* 2002;80:235–55.
- [47] Bathe KJ, Lee PS, Hiller JF. Towards improving the MITC9 shell element. *Comput Struct* 2003;81:477–89.
- [48] Ko Y, Lee PS. A 6-node triangular solid-shell element for linear and nonlinear analysis. *Int J Numer Meth Eng* 2017;111:1203–30.
- [49] Cen S, Chen XM, Fu XR. Quadrilateral membrane element family formulated by the quadrilateral area coordinate method. *Comput Methods Appl Mech Eng* 2007;196:4337–53.
- [50] Jung J, Jun H, Lee PS. Self-updated four-node finite element using deep learning. *Comput Mech* 2022;69:23–44.
- [51] Sze KY, Liu XH, Lo SH. Popular benchmark problems for geometric nonlinear analysis of shells. *Finite Elem Anal Des* 2004;40:1551–69.
- [52] Macneal RH, Harder RL. A proposed standard set of problems to test finite element accuracy. *Finite Elem Anal Des* 1985;1:3–20.